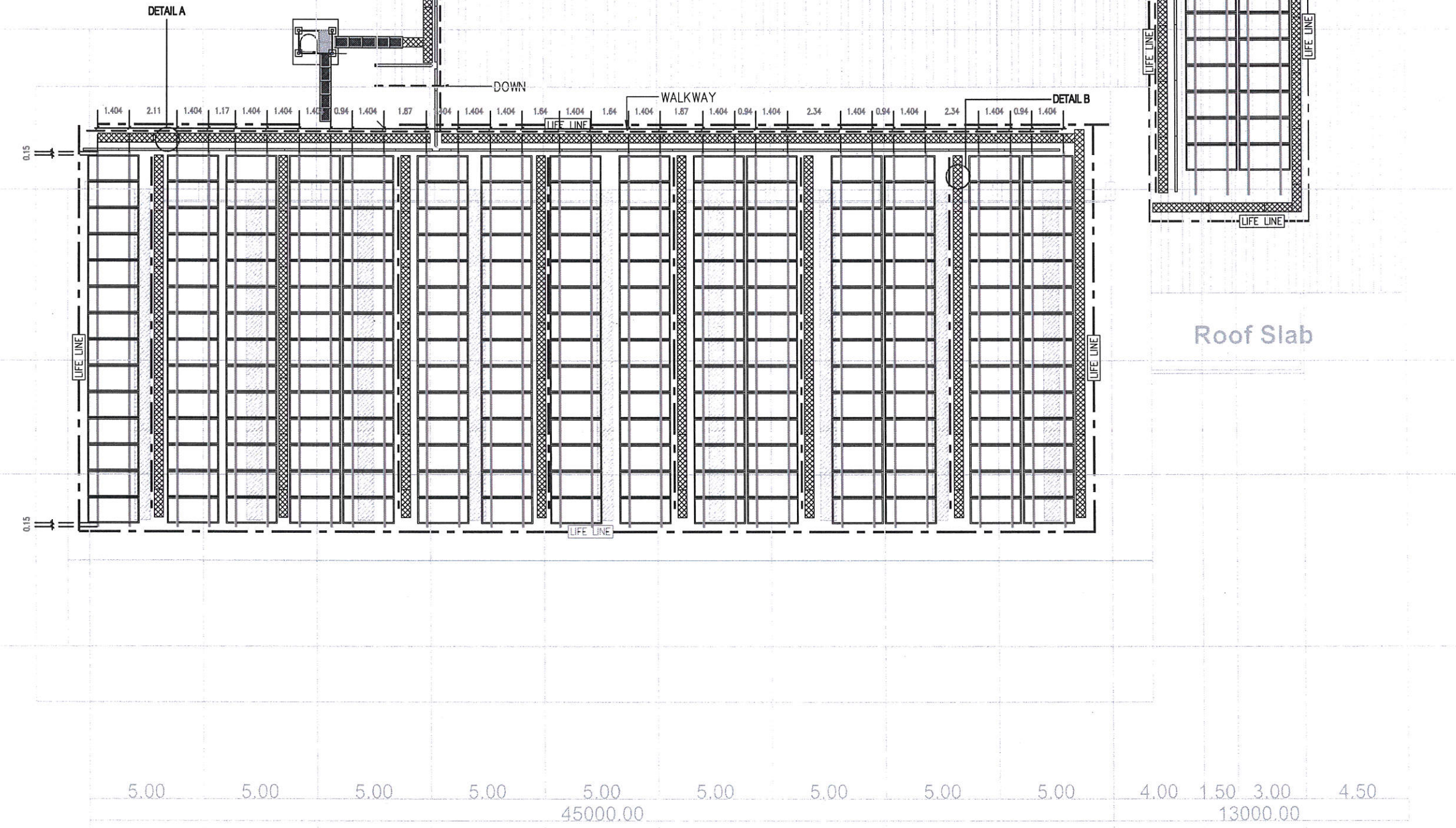
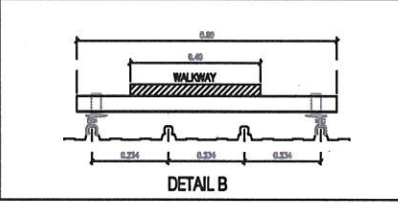


4	05/06/2023	AS-BUILT				
3	31/05/2023	Final for Construction				
2	30/05/2023	Issued for Approval				
1	24/05/2023	Issued for Approval				
0	11/10/2022	Issued for Approval				
Rev.	Date	Description				
Owner : ELECTRICITY GENERATING AUTHORITY OF THAILAND						
Contractor : RATCH Group Public Company Limited						
ARCHITECT : -		SUBMITTED		SAI NOI PV AND BESS PROJECT		
ENGINEER : -						
ENGINEER : -		CONCURRED		Drawing name Plan view for PV mounting, walkway and ladder for each building.		
DRAWN : -						
CHECKED : -						
APPROVED		REPLACING DWG. NO. -		DWG. NO. SNPB-D-GRC-WTA-001	REV NO. 4	SHEET NO. 1/5
.....		DATE				


Rev. 10723



อาคาร ท.0007

Part
of the of the
NW. 10773

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN

NOTE.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
0	FOR APPROVAL	22/06/2023	KWANPONG	KWANPONG	SURASAK
1	FOR APPROVAL	24/06/2023	PhumthaiA	MertholaA	MertholaA
2	FOR APPROVAL	26/06/2023	PhumthaiA	MertholaA	MertholaA
3	FOR CONSTRUCTION	27/06/2023	PhumthaiA	MertholaA	MertholaA
4	AS-BUILT	02/06/2023	PhumthaiA	MertholaA	MertholaA

DESCRIPTION :	
---------------	--

Plan View for PV Mounting, Walkway and
Ladder for Each Building
(T.0007)

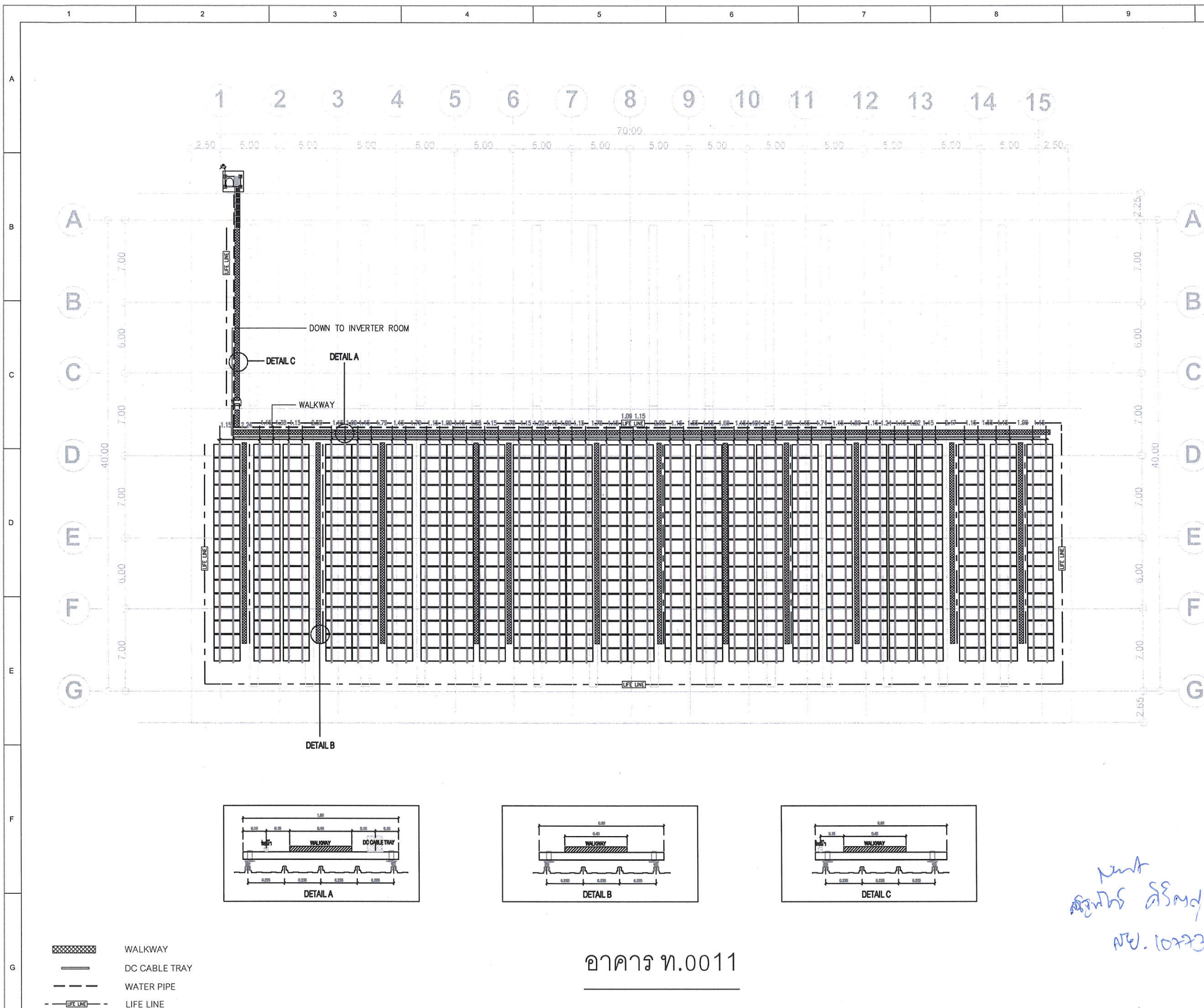
DWG NO. SNPB-D-GRC-WTA-001

JOB NO.	
---------	--

SHEET : 1 / 4

SIZE : A3	S
-----------	---

SCALE : 1:100



PROJECT NAME :	
----------------	--

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkrui-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



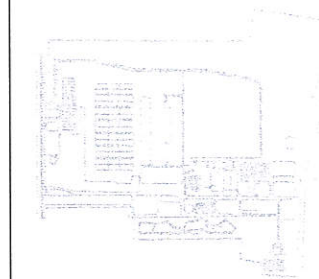
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD
0	FOR APPROVAL	24/06/2022	KHANFONG	KHANFONG	SUEASA
1	FOR APPROVAL	24/06/2022	Phumduaj	MerthA	MerthA
2	FOR APPROVAL	20/06/2022	Phumduaj	MerthA	MerthA
3	FOR CONSTRUCTION	21/06/2022	Phumduaj	MerthA	MerthA
4	AS-BUILT	01/08/2022	Phumduaj	MerthA	MerthA

	DESCRIPTION :

Plan View for PV Mounting, Walkway and
Ladder for Each Building
(T.0011)

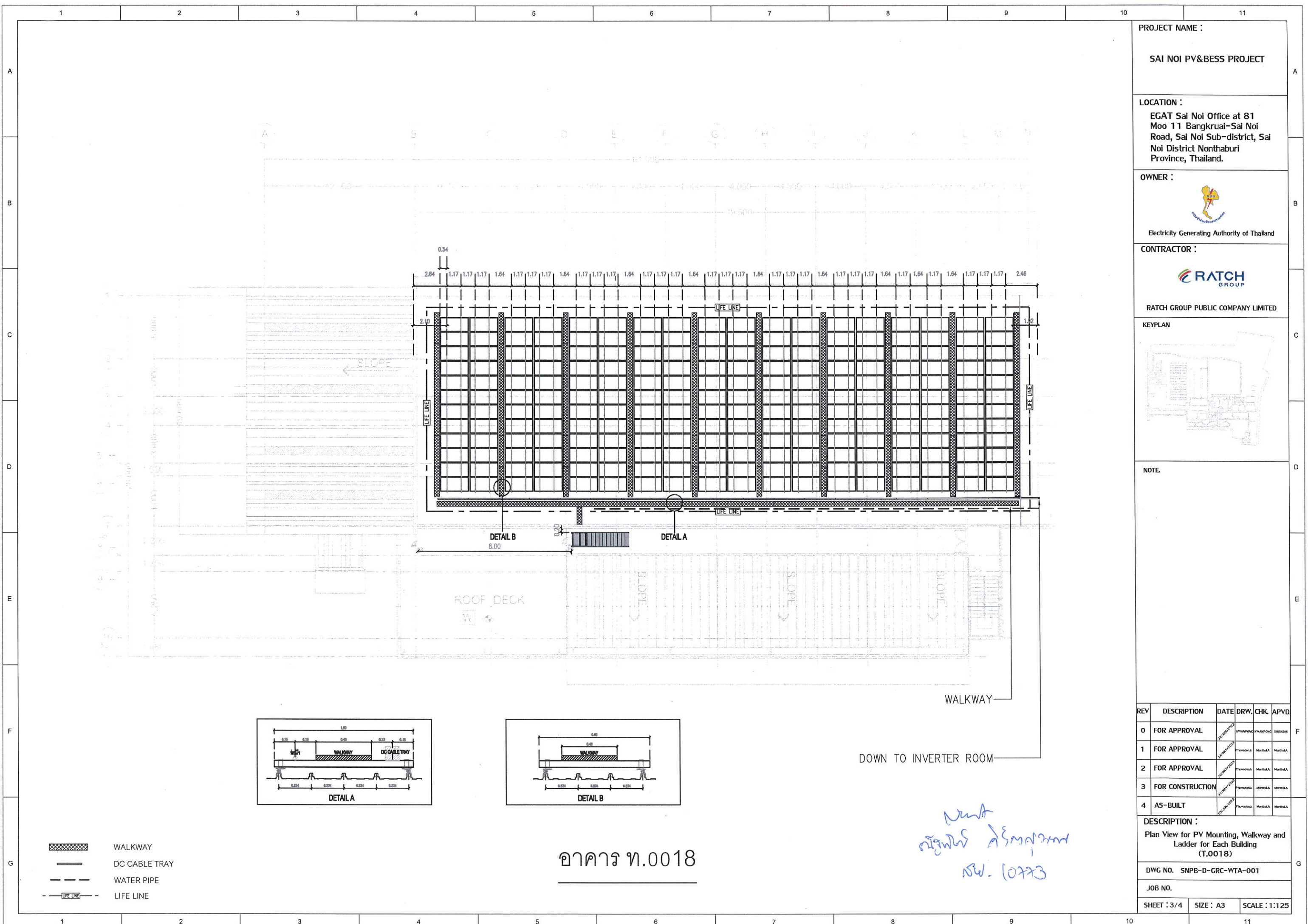
DWG NO. SNPB-D-GRC-WTA-001

	JOB NO.
--	---------

SHEET : 2/4

SIZE : A3	SCALE : 1:150
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อาคาร ท.0011



PROJECT NAME :

SAI NOI PV&BESS PROJECT

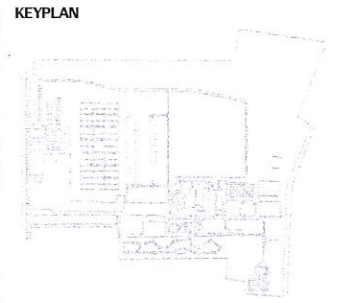
LOCATION :
EGAT Sai Noi Office at 81
Moo 11 Bangkrual-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :

Electricity Generating Authority of Thailand

CONTRACTOR :

RATCH GROUP PUBLIC COMPANY LIMITED



NOTE

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
0	FOR APPROVAL	23/04/2023	Wanpang	Wanpang	Surasak
1	FOR APPROVAL	24/04/2023	Phonadok	ManthadA	ManthadA
2	FOR APPROVAL	25/04/2023	Phonadok	ManthadA	ManthadA
3	FOR CONSTRUCTION	25/04/2023	Phonadok	ManthadA	ManthadA
4	AS-BUILT	02/05/2023	Phonadok	ManthadA	ManthadA

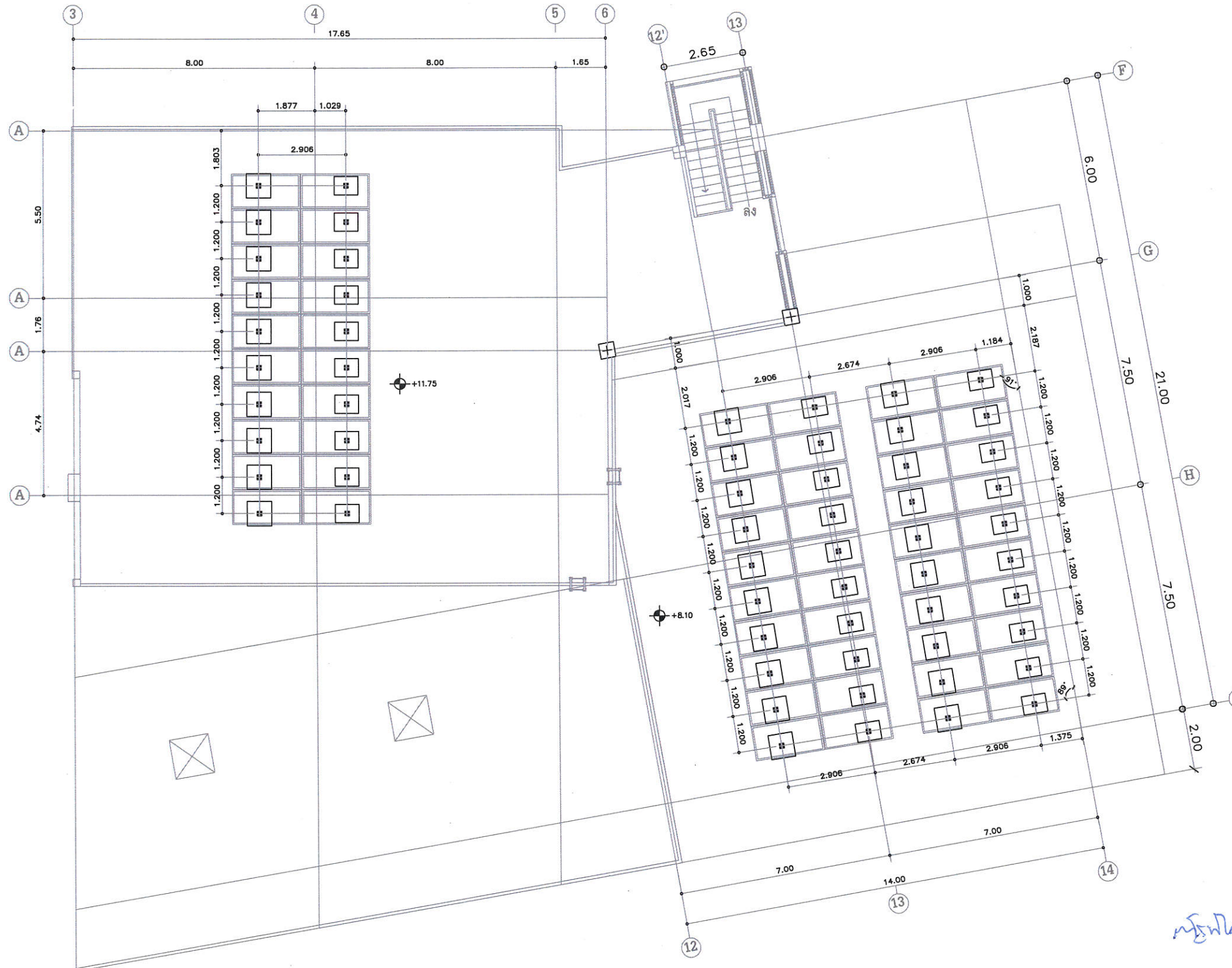
DESCRIPTION :
Plan View for PV Mounting, Walkway and
Ladder for Each Building
(T.0018)

DWG NO. SNPB-D-GRC-WTA-001

JOB NO.

SHEET : 3/4 SIZE : A3 SCALE : 1:125

หน้า
ฉบับร่าง
วันที่ 10/7/23



อาคาร ท.0020

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkrual-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
0	FOR APPROVAL	22 MAR 2020	KHANPONG KHANPONG	SURASAK	
1	FOR APPROVAL	24 MAR 2020	Phumadit	MordulA	MordulA
2	FOR APPROVAL	30 MAR 2020	Phumadit	MordulA	MordulA
3	FOR CONSTRUCTION	21 MAR 2020	Phumadit	MordulA	MordulA
4	AS-BUILT	02 APR 2020	Phumadit	MordulA	MordulA

DESCRIPTION :

Plan View for PV Mounting, Walkway and
Ladder for Each Building
(T.0020)

DWG NO. SNPB-D-GRC-WTA-001

JOB NO.

SHEET : 4/4

SIZE : A3

SCALE : 1:100

Handwritten signature and date: 10/7/23



ใช้ในการรับรองแบบสำหรับ Plan View for PV Mounting,
Walkway and ladder for each building

โครงการ SAI NOI PV AND BESS PROJECT

โรงไฟฟ้าไทรน้อย ตำบลไทรน้อย อำเภอไทรน้อย จังหวัดนนทบุรี

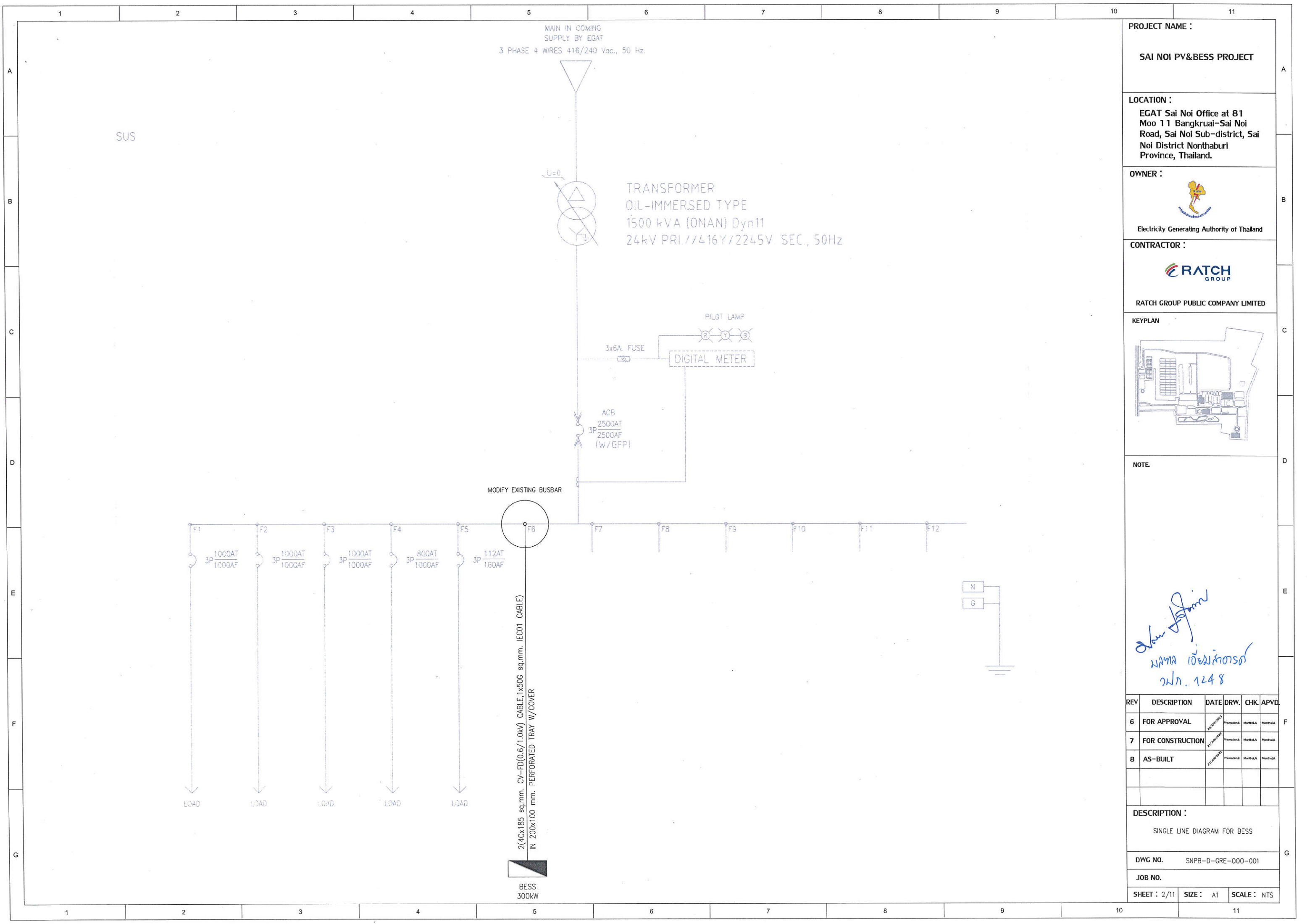
pent
(ณัฐพัชร ศรีคงสุวรรณ)
สย. 10773



8	23/06/2023	As-Built				
7	21/06/2023	Final for Construction				
6	26/04/2023	Issued for Approval				
Rev.	Date	Description				
Owner : ELECTRICITY GENERATING AUTHORITY OF THAILAND						
Contractor : RATCH Group Public Company Limited						
ARCHITECT : -		SUBMITTED	SAI NOI PV AND BESS PROJECT			
ENGINEER : -						
ENGINEER : -		CONCURRED	Drawing name Single Line Diagram			
DRAWN : -						
CHECKED : -						
APPROVED		DATE.....	REPLACING DWG. NO. -	DWG. NO. SNPB-D-GRE- 000-001	REV NO. 8	SHEET NO. 1/12


 นามานี เดชะวิจิตร
 ๓๓. ๑๒๔๘

1	2	3	4	5	6	7	8	9	10	11
ELECTRICAL ABBREVIATIONS		ELECTRICAL ABBREVIATIONS		DISTRIBUTION PANELBOARD & DEVICES		FIRE ALARM SYSTEM		PROJECT NAME :		
ABB. TERM		ABB. TERM		SYMBOLS DESCRIPTION		SYMBOLS DESCRIPTION		SAI NOI PV&BESS PROJECT		
WHERE ABBREVIATIONS ARE USED IN THE DOCUMENTS AND NOT LISTED IN THE FOLLOWING, REFER TO ANSI/IEEE STANDARD 100 AND FOR MECHANICAL EQUIPMENT REFER TO THE MECHANICAL SYSTEMS ABBREVIATIONS SCHEDULE.		WHERE ABBREVIATIONS ARE USED IN THE DOCUMENTS AND NOT LISTED IN THE FOLLOWING, REFER TO ANSI/IEEE STANDARD 100 AND FOR MECHANICAL EQUIPMENT REFER TO THE MECHANICAL SYSTEMS ABBREVIATIONS SCHEDULE.		WITH 2 NO AND 1 NC AUXILIARY CONTACTS WITH 2 NO AND 1 NC AUXILIARY CONTACTS (SHOWN SMALLER THAN THE MAIN CONTACTS)		WITH 2 NO AND 1 NC AUXILIARY CONTACTS WITH 2 NO AND 1 NC AUXILIARY CONTACTS (SHOWN SMALLER THAN THE MAIN CONTACTS)		LOCATION :		
A AMPERE		NTS NOT TO SCALE		SHUNT TRIP		SHUNT TRIP		EGAT Sai Noi Office at 81		
AC ALTERNATING CURRENT		OH OVERHEAD		UNDERVOLTAGE TRIPPING DEVICE OR RELAY		UNDERVOLTAGE TRIPPING DEVICE OR RELAY		Moo 11 Bangkruai-Sai Noi		
ACB AIR CIRCUIT BREAKER		OL OVERLOAD		AMP METER		AMP METER		Road, Sai Noi Sub-district, Sai		
ACC ACCESSORIES		P POLE		VOLTMETER		VOLTMETER		Noi District Nonthaburi		
AF FRAME SIZE OF CIRCUIT BREAKER IN AMPERE		PA PUBLIC ADDRESS		PILOT LAMP, R=RED, Y=YELLOW, B=BLUE, G=GREEN, C=CLEAR, etc.		PILOT LAMP, R=RED, Y=YELLOW, B=BLUE, G=GREEN, C=CLEAR, etc.		Province, Thailand.		
AFF ABOVE FINISHED FLOOR		PABX PRIVATE AUTOMATIC BRANCH EXCHANGE		EARTH FAULT RELAY		EARTH FAULT RELAY		OWNER :		
AFG ABOVE FINISHED GRADE		PEA PROVINCIAL ELECTRICITY AUTHORITY		TIME DELAY RELAY		TIME DELAY RELAY		Electricity Generating Authority of Thailand		
AHU AIR HANDLING UNIT		PB, EPB PANELBOARD, ESSENTIAL PANELBOARD		TEMPERATURE RELAY		TEMPERATURE RELAY		CONTRACTOR :		
AL ALUMINIUM		PH PHASE		MONITOR COMPONENT OF GROUND FAULT SENSOR		MONITOR COMPONENT OF GROUND FAULT SENSOR		RATCH GROUP		
AT AMPERE TRIP OF CIRCUIT BREAKER		PT POTENTIAL TRANSFORMER		MOTOR OPERATED MECHANISM		MOTOR OPERATED MECHANISM		RATCH GROUP PUBLIC COMPANY LIMITED		
ATS AUTOMATIC TRANSFER SWITCH		PVC POLY-VINYL CHLORIDE		PHASE FAILURE RELAY		PHASE FAILURE RELAY		KEYPLAN		
AWG AMERICAN WIRE GAUGE		RCD RESIDUAL CURRENT DEVICE		POWER FACTOR CONTROLLER		POWER FACTOR CONTROLLER		NOTE		
A/C AIR CONDITION		RC REMOTE CONTROL		GROUND FAULT SENSOR AND RELAY, USING WINDOW TYPE CURRENT TRANSFORMER WITH TIME DELAY		GROUND FAULT SENSOR AND RELAY, USING WINDOW TYPE CURRENT TRANSFORMER WITH TIME DELAY		REV DESCRIPTION DATE DRW. CHK. APVD.		
BC BARE COPPER		RCP REFLECTED CEILING PLAN		DIGITAL MULTIMETER		DIGITAL MULTIMETER		6 FOR APPROVAL 20/06/2023 Phasakorn Muthakul Muthakul		
BLDG. BUILDING		REF REFERENCE		AMP METER SELECTOR SWITCH		AMP METER SELECTOR SWITCH		7 FOR CONSTRUCTION 21/06/2023 Phasakorn Muthakul Muthakul		
CB CIRCUIT BREAKER		RL REFERENCE LEVEL		VOLT METER SELECTOR SWITCH		VOLT METER SELECTOR SWITCH		8 AS-BUILT 23/06/2023 Phasakorn Muthakul Muthakul		
CCT CIRCUIT		RMS ROOT MEAN SQUARE		FREQUENCY METER		FREQUENCY METER		DESCRIPTION :		
CCTV CLOSED CIRCUIT TELEVISION		RSC RIGID STEEL CONDUIT		POWER FACTOR METER		POWER FACTOR METER		LEGEND AND ABBREVIATION		
CDU CONDENSING UNIT		SN SANITARY		KILOWATT - HOUR METER		KILOWATT - HOUR METER		DWG NO. SNPB-D-GRE-000-001		
CLG CEILING		SPN SINGLE POLE AND NEUTRAL		KILOVAR METER		KILOVAR METER		JOB NO.		
CM CENTIMETERS		ST SHUNT TRIP		HOUR METER		HOUR METER		SHEET : 1/11		
CO CONDUIT ONLY		SW SWITCH		THERMAL OVERLOAD TRIPPING DEVICE		THERMAL OVERLOAD TRIPPING DEVICE		SIZE : A1		
COMM. COMMUNICATION SYSTEM		SWGR SWITCHGEAR		MAGNETIC OVERLOAD TRIPPING DEVICE		MAGNETIC OVERLOAD TRIPPING DEVICE		SCALE : NTS		
CP CONTROL PANEL		SYM SYMMETRICAL		NORMALLY OPEN AUXILIARY CONTACT (NO)		NORMALLY OPEN AUXILIARY CONTACT (NO)				
CPU CENTRAL PROCESSING UNIT		TB TERMINAL BOX		NORMALLY CLOSED AUXILIARY CONTACT (NC)		NORMALLY CLOSED AUXILIARY CONTACT (NC)				
CT CURRENT TRANSFORMER		TBA TO BE ADVISED		CONTACTOR CONTACT		CONTACTOR CONTACT				
CU COPPER		TBC TO BE CONFIRMED		PUSH BUTTON NO		PUSH BUTTON NO				
DB, EDB DISTRIBUTION BOARD, ESSENTIAL DISTRIBUTION BOARD		TC TERMINAL CABINET FOR TELEPHONE, SIGNAL, ETC.		PUSH BUTTON NC		PUSH BUTTON NC				
DC DIRECT CURRENT		TEL TELEPHONE		TIME - DELAY CONTACT		TIME - DELAY CONTACT				
DEG DEGREE		TR TRANSFORMER		INTERLOCKING FUNCTION		INTERLOCKING FUNCTION				
DIA DIAMETER		TPN TRIPLE POLE AND NEUTRAL		CAPACITOR BANK		CAPACITOR BANK				
DIM DIMMER		TOT TELEPHONE AUTHORITY OF THAILAND		BATTERY		BATTERY				
DISC DISCONNECT SWITCH/ISOLATOR		TV TELEVISION		BUSDUCT PLUG IN UNIT WITH FUSIBLE SWITCH (RATING AS SPECIFIED)		BUSDUCT PLUG IN UNIT WITH FUSIBLE SWITCH (RATING AS SPECIFIED)				
DN DOWN		TYP. TYPICAL		BUSDUCT PLUG IN UNIT WITH INTERNAL MCCB (RATING AS SPECIFIED)		BUSDUCT PLUG IN UNIT WITH INTERNAL MCCB (RATING AS SPECIFIED)				
DWG DRAWING		UG UNDERGROUND		MAGNETIC CONTACTOR		MAGNETIC CONTACTOR				
° DEGREE CELSIUS		UL UNDERWRITER'S LABORATORY		MOTOR CONTROL CENTER		MOTOR CONTROL CENTER				
° DEGREE FAHRENHEIT		UPS UNINTERRUPTIBLE POWER SUPPLY		CONNECTED TO GROUND ROD		CONNECTED TO GROUND ROD				
dB DECIBEL		V VOLTS		MAIN DISTRIBUTION BOARD		MAIN DISTRIBUTION BOARD				
E EMERGENCY		VA VOLT-AMPERE		ESSENTIAL MAIN DISTRIBUTION BOARD		ESSENTIAL MAIN DISTRIBUTION BOARD				
EGAT ELECTRICITY GENERATING AUTHORITY OF THAILAND		VS VOLT-SELECTOR SWITCH		DISTRIBUTION BOARD		DISTRIBUTION BOARD				
EL ELEVATION		W WATTS		ESSENTIAL DISTRIBUTION BOARD		ESSENTIAL DISTRIBUTION BOARD				
ELCB EARTH LEAKAGE CIRCUIT BREAKER		WP WEATHERPROOF		PANELBOARD		PANELBOARD				
EMT ELECTRICAL METALLIC TUBING		WWTP WASTE WATER TREATMENT PLANT		ESSENTIAL PANELBOARD		ESSENTIAL PANELBOARD				
EXP EXPLOSION PROOF		Y WYE		MECHANICAL CONTROL CENTER		MECHANICAL CONTROL CENTER				
EXT EXISTING		Z IMPEDANCE		UNINTERRUPTIBLE POWER SUPPLY		UNINTERRUPTIBLE POWER SUPPLY				
F FUSE OR FUSED				LIGHTING CONTROL UNIT		LIGHTING CONTROL UNIT				
FA FIRE ALARM				GENERATOR CONTROL PANEL		GENERATOR CONTROL PANEL				
FCP FIRE ALARM CONTROL PANEL				DISCONNECT SWITCH WITH FUSE		DISCONNECT SWITCH WITH FUSE				
FL FLOOR				SERVICE ENTRANCE CAP		SERVICE ENTRANCE CAP				
F OR FDR FEEDER				MANHOLE		MANHOLE				
FRC FIRE RESISTANT CABLE				TELEPHONE MANHOLE		TELEPHONE MANHOLE				
FVNR FULL VOLTAGE NON REVERSING				HANDHOLE		HANDHOLE				
GEN GENERATOR				PULL BOX		PULL BOX				
HDPE HIGH DENSITY POLYETHYLENE				MOLDED CASE CIRCUIT BREAKER WITH ENCLOSURE		MOLDED CASE CIRCUIT BREAKER WITH ENCLOSURE				
G GROUND				TRANSFORMER STATION		TRANSFORMER STATION				
G/L GRID LINE				CONCRETE POLE		CONCRETE POLE				
GFI GROUND FAULT CIRCUIT INTERRUPTER				SPN/TPN SAFETY SWITCH		SPN/TPN SAFETY SWITCH				
HID HIGH INTENSITY DISCHARGE				SPN/TPN SAFETY SWITCH WITH FUSE		SPN/TPN SAFETY SWITCH WITH FUSE				
HP HORSE-POWER										
HV HIGH VOLTAGE										
H OR HT HEIGHT										
RSC GALVANIZED RIGID STEEL CONDUIT										
HZ HERTZ										
IC INTERRUPTING CAPACITY (BREAKER CAPACITY)										
IL INVERT LEVEL										
IMC INTERMEDIATE METALLIC CONDUIT										
IP INGRESS PROTECTION										
J JUNCTION BOX										
K KILO, THOUSAND										
LBS LOAD BREAK SWITCH										
LCU LIGHTING CONTROL UNIT										
LV LOW VOLTAGE										
M METERS										
MATV MASTER ANTENNA TELEVISION										
MCB MINIATURE CIRCUIT BREAKER										
MCC MOTOR CONTROL CENTER										
MCCB MOLDED CASE CIRCUIT BREAKER										
MCM THOUSAND CIRCULAR MILS										
MDB, EMDB MAIN DISTRIBUTION BOARD, ESSENTIAL MAIN DISTRIBUTION BOARD										
MDF MAIN DISTRIBUTION FRAME FOR TELEPHONE SYSTEM										
MEA METROPOLITAN ELECTRICITY AUTHORITY										
MH MANHOLE										
MISC MISCELLANEOUS										
MM MILLIMETERS										
MOA MANUAL-OFF-AUTOMATIC										
MT MOUNT										
NA NOT APPLICABLE										
NC NORMALLY CLOSED										
NEC NATIONAL ELECTRICAL CODE										
NF NON FUSED										
NFPA NATIONAL FIRE PROTECTION ASSOCIATION										
NIC NOT IN CONTRACT										
NL NIGHT LIGHT										
NO NORMALLY OPEN										



PROJECT NAME :

SAI NOI PV&BESS PROJECT

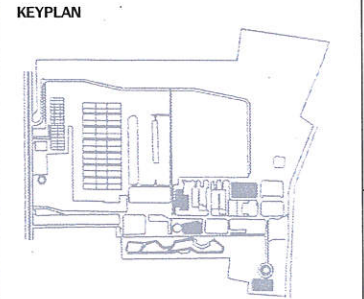
LOCATION :
EGAT Sai Noi Office at 81
Moo 11 Bangkrui-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :

Electricity Generating Authority of Thailand

CONTRACTOR :

RATCH GROUP PUBLIC COMPANY LIMITED



NOTE

Handwritten signature and text:
นาย วัฒนวิทย์
พ.ร. 1248

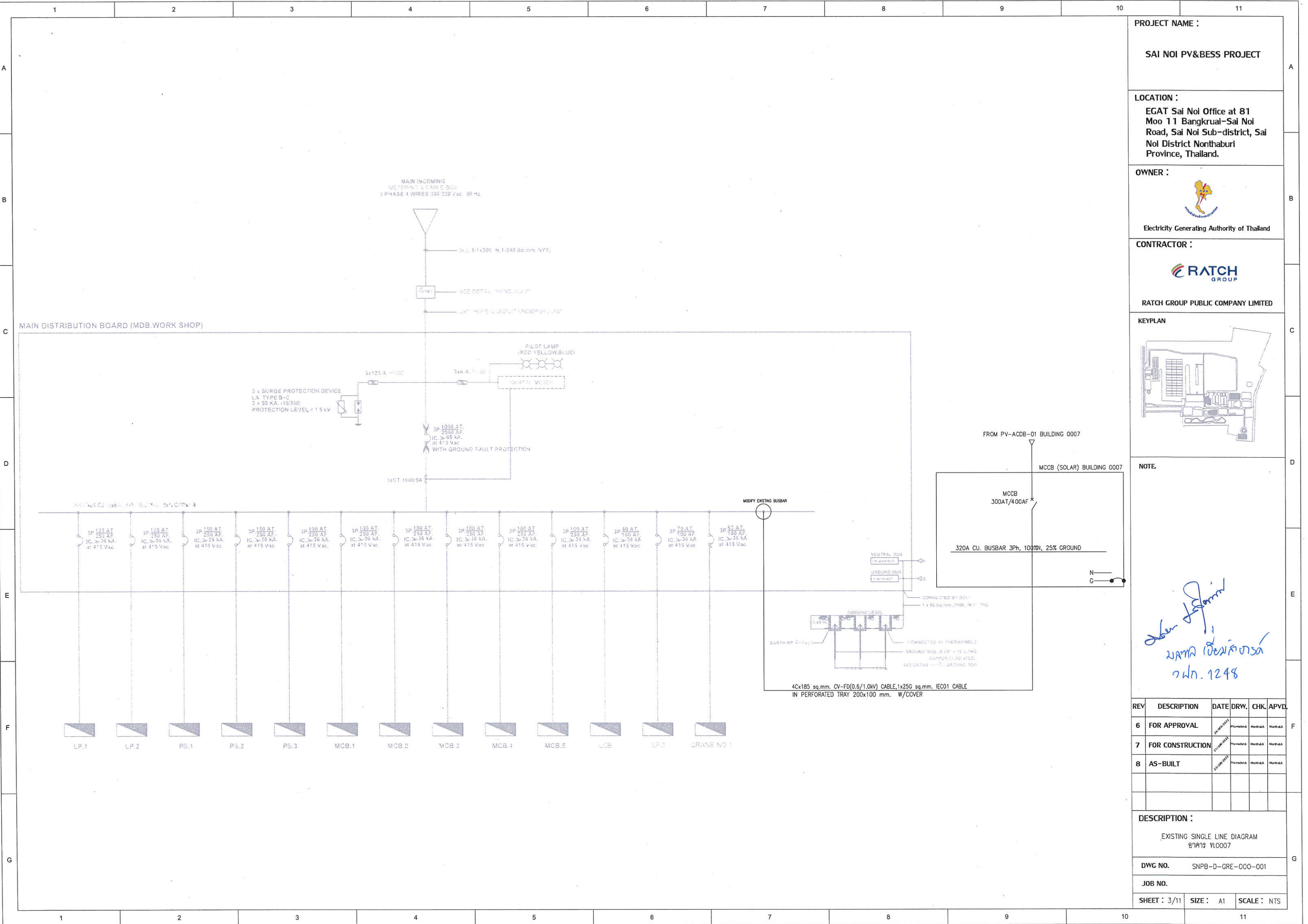
REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
6	FOR APPROVAL	11/06/2023	Phumchok	Montthak	Montthak
7	FOR CONSTRUCTION	11/06/2023	Phumchok	Montthak	Montthak
8	AS-BUILT	21/06/2023	Phumchok	Montthak	Montthak

DESCRIPTION :
SINGLE LINE DIAGRAM FOR BESS

DWG NO. SNPB-D-GRE-000-001

JOB NO.

SHEET : 2/11 SIZE : A1 SCALE : NTS



PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkrui-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



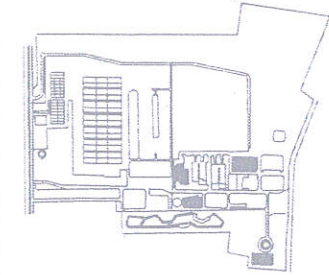
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE

Handwritten note in Thai script:
งานติดตั้งระบบไฟฟ้า
วันที่ 12/48

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
6	FOR APPROVAL	24/04/2023	Phumadok	Montida	Montida
7	FOR CONSTRUCTION	21/04/2023	Phumadok	Montida	Montida
8	AS-BUILT	21/04/2023	Phumadok	Montida	Montida

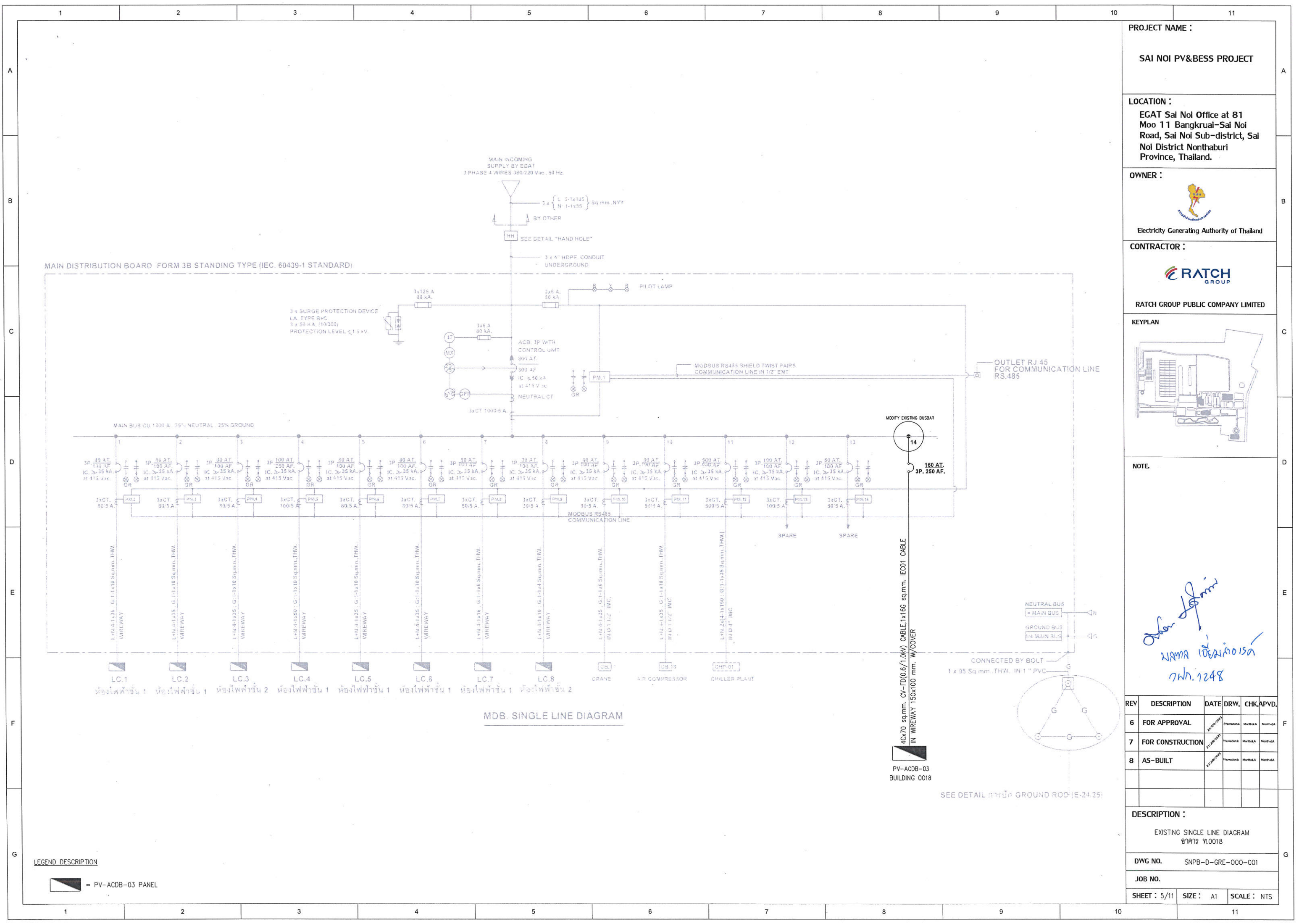
DESCRIPTION :

EXISTING SINGLE LINE DIAGRAM
ฉบับที่ 110007

DWG NO. SNPB-D-GRE-000-001

JOB NO.

SHEET : 3/11 SIZE : A1 SCALE : NTS



PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :

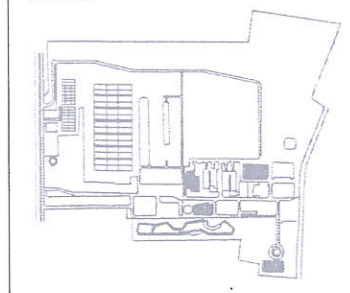


CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Handwritten signature and date: 12/24/25

REV	DESCRIPTION	DATE	DRW.	CHK	APVD.
6	FOR APPROVAL	24/12/25	Phonarak	Monti-A	Monti-A
7	FOR CONSTRUCTION	21/12/25	Phonarak	Monti-A	Monti-A
8	AS-BUILT	23/12/25	Phonarak	Monti-A	Monti-A

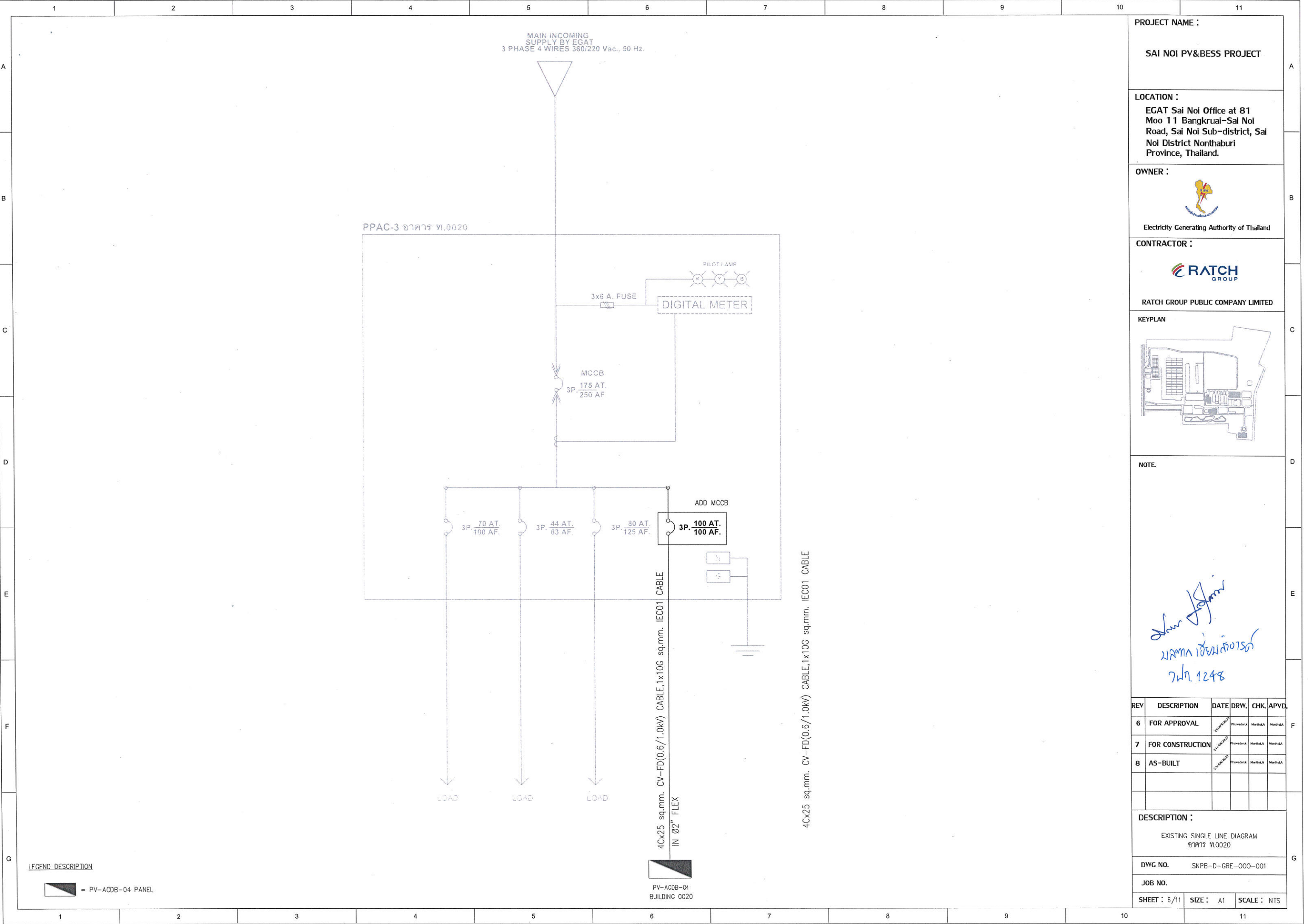
DESCRIPTION :

EXISTING SINGLE LINE DIAGRAM
ฉบับที่ 00018

DWG NO. SNPB-D-GRE-000-001

JOB NO.

SHEET : 5/11 SIZE : A1 SCALE : NTS



PROJECT NAME :

SAI NOI PV&BESS PROJECT

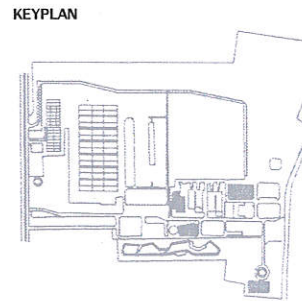
LOCATION :
EGAT Sai Noi Office at 81
Moo 11 Bangkrui-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :

Electricity Generating Authority of Thailand

CONTRACTOR :

RATCH GROUP PUBLIC COMPANY LIMITED



NOTE

San Jom
นางสาวเจษฎาภรณ์
พ.พ. 1248

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
6	FOR APPROVAL	20/08/2023	Phumabok	MontiA	MontiA
7	FOR CONSTRUCTION	21/08/2023	Phumabok	MontiA	MontiA
8	AS-BUILT	22/08/2023	Phumabok	MontiA	MontiA

DESCRIPTION :
EXISTING SINGLE LINE DIAGRAM
อาคาร พ.0020

DWG NO. SNPB-D-GRE-000-001

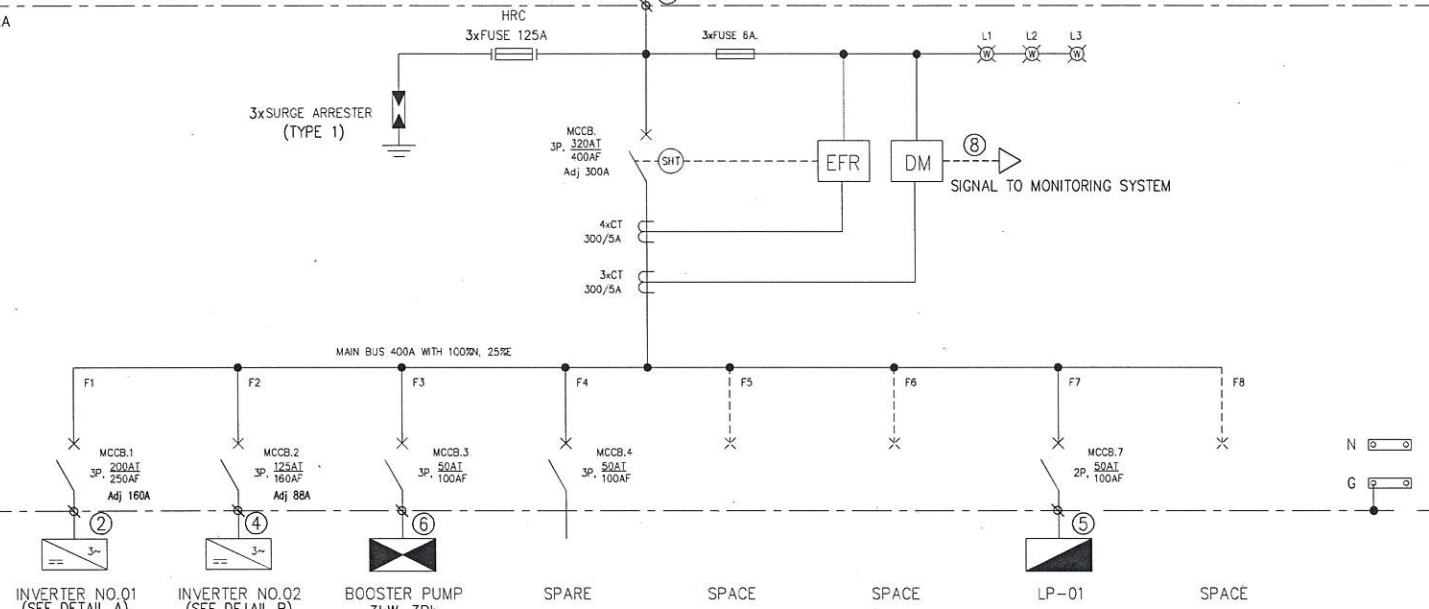
JOB NO.

SHEET : 6/11 SIZE : A1 SCALE : NTS

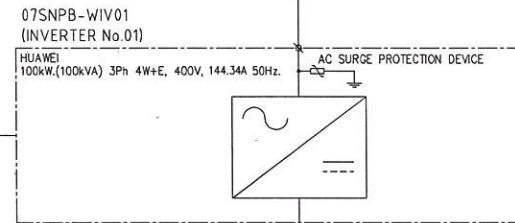
07SNPB-WAC01
(PV-ACDB-01 BUILDING 0007)

ALL CB IC > 30 kA
AT 415 VAC
IP41

TO MCCB (SOLAR) BUILDING 0007

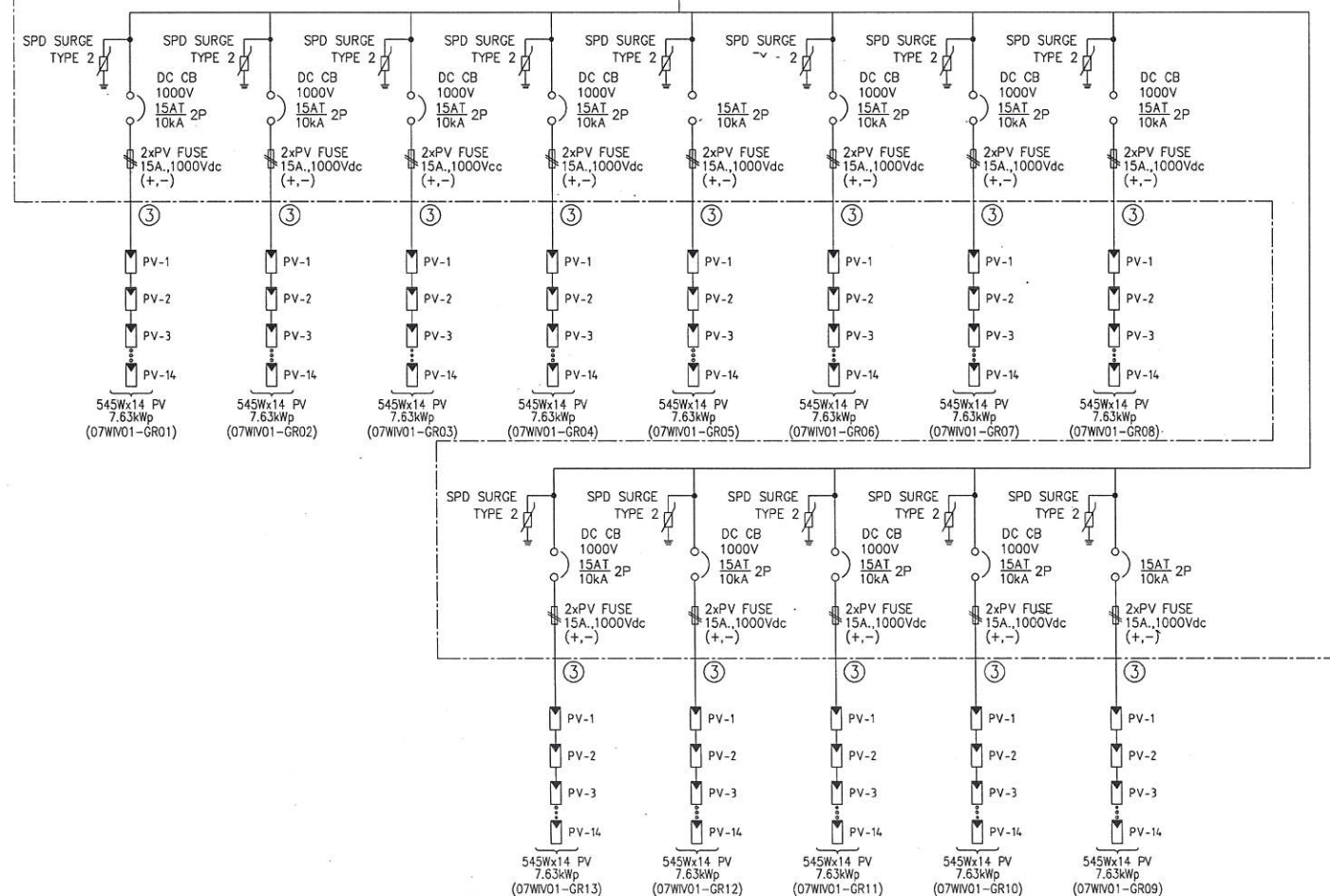


07SNPB-WAC01
(PV-ACDB-01 BUILDING 0007)



- NOTES :
1. MAIN MCCB 3P 320AT/400AF (ADJUST TO 300AT) IS PROVIDED WITH CLOSING /TRIPPING COIL UNIT FOR RECEIVING SIGNAL FROM PROTECTION RELAYS AND TIMER.
 2. INVERTER NO.01 USES A TOTAL OF 10 MPPT.
 3. INVERTER NO.02 USES A TOTAL OF 6 MPPT.
 4. MORE DETAIL OF MPPT CONNECTION SHALL REFER TO SNPB-D-GRE-WAQ-009 CABLE SCHEDULE.

07SNPB-WDC01
(DC PANEL BOX 01 BUILDING 0007)



13 STRINGS/182 PV PANELS (99.190kWp)

DETAIL A - INVERTER 100kW

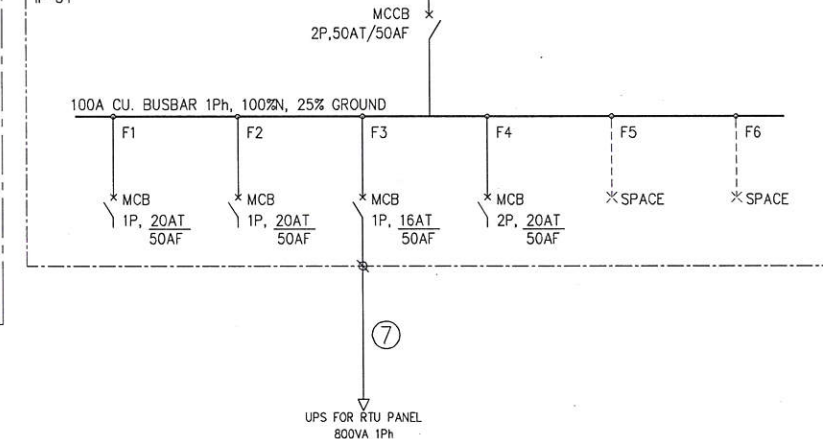
LEGEND DESCRIPTION

- (SHT) = SHUNT TRIP
(DM) = DIGITAL METER
(EFR) = EARTH FAULT RELAY

07SNPB-WAC01
(PV-ACDB-01 BUILDING 0007)

07SNPB-WLP01
(LP-01)

Isc > 25kA.
IP 54



DETAIL LP-01

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



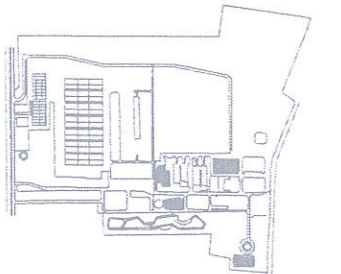
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

CABLE KEY NOTES :

CODE	
①	4Cx185 sq.mm. CV-FD(0.6/1.0kV) CABLE, 1x25G sq.mm. IEC01 CABLE IN PERFORATED TRAY 200x100 mm. W/COVER
②	4Cx95 sq.mm. CV-FD(0.6/1.0kV), G1x16 sq.mm. IEC01 CABLE IN PERFORATED TRAY 200x100 mm. W/COVER
③	2x6 sq.mm. PV1-F CABLE IN PERFORATED TRAY 100x100 mm. W/COVER
④	4Cx35 sq.mm. CV-FD(0.6/1.0kV), G1x16 sq.mm. IEC01 CABLE IN PERFORATED TRAY 200x100 mm. W/COVER
⑤	2-10/6G IEC01 IN Ø3/4" FLEX
⑥	4C-10 sq.mm. CV CABLE, G1x6 sq.mm. IEC01 CABLE IN Ø32 mm. HDPE PIPE
⑦	2-10/6G IEC01 IN Ø1/2" FLEX
⑧	2 CORES 16AWG SHIELD OUTDOOR TYPE IN Ø1/2" FLEX
⑨	13(1x6 sq.mm. PV1-F CABLE) IN Ø1 1/4" FLEX (POSITIVE)
	13(1x6 sq.mm. PV1-F CABLE) IN Ø1 1/4" FLEX (NEGATIVE)

Signature
นาย อดิศักดิ์ อดิศักดิ์
24.12.2568

REV	DESCRIPTION	DATE	DRW	CHK	APVD
6	FOR APPROVAL	21/12/2023	Promsarak	Montthak	Montthak
7	FOR CONSTRUCTION	21/12/2023	Promsarak	Montthak	Montthak
8	AS-BUILT	21/12/2023	Promsarak	Montthak	Montthak

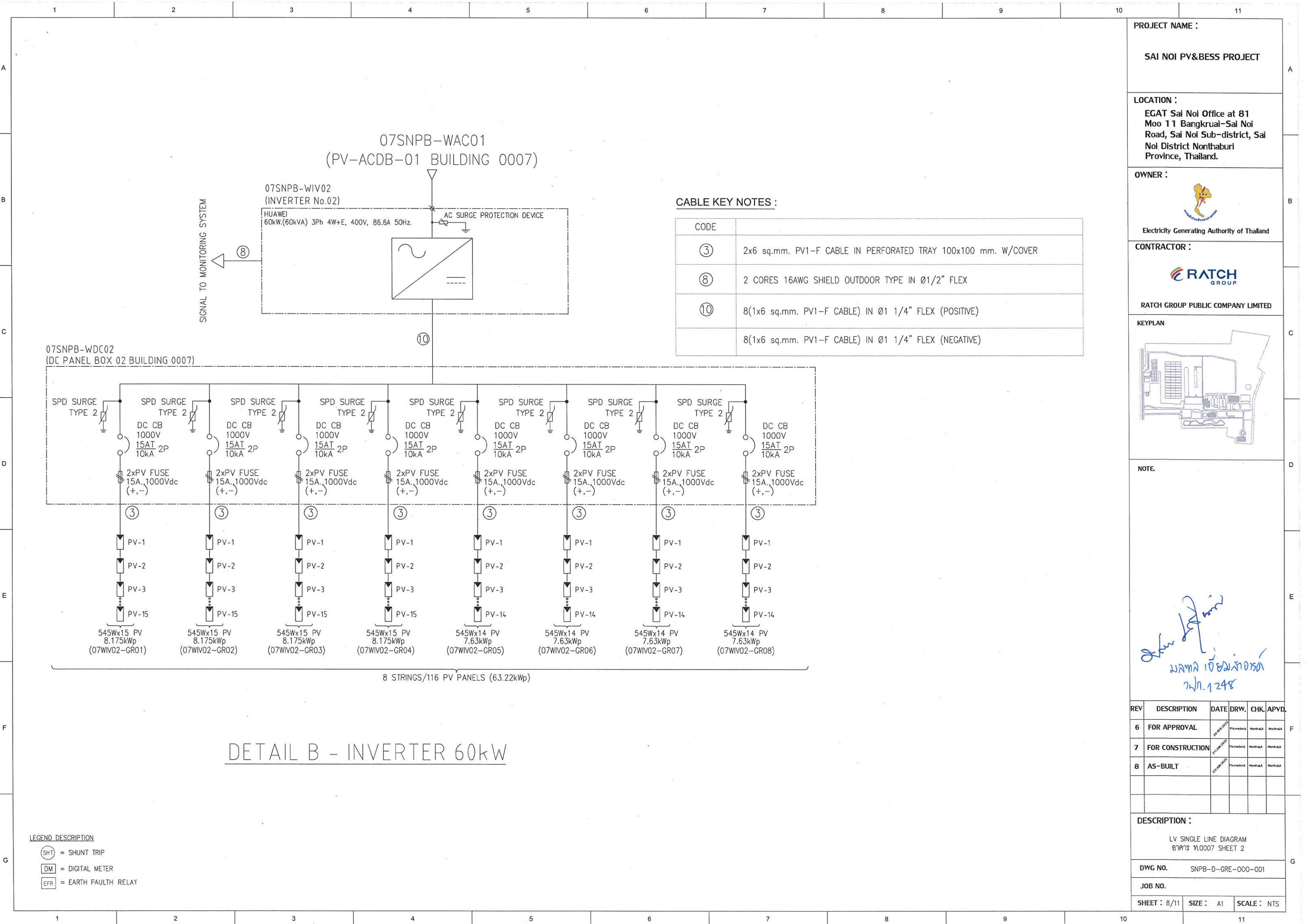
DESCRIPTION :

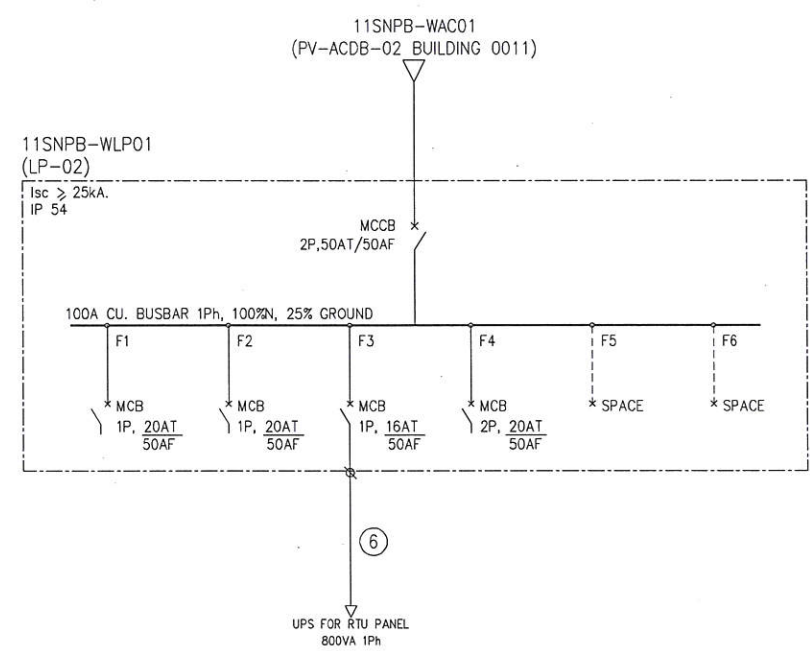
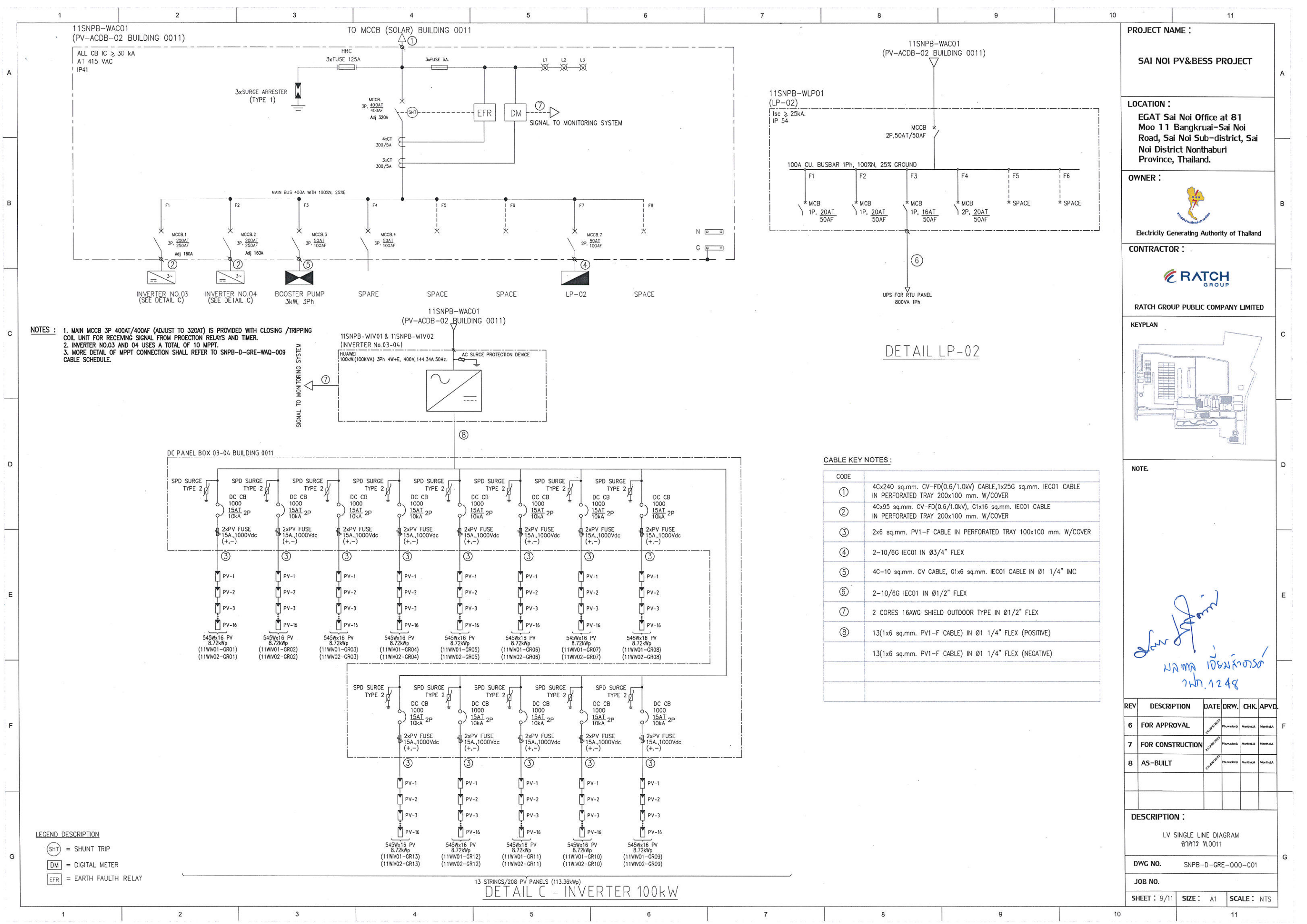
LV SINGLE LINE DIAGRAM
07WV01 Y0007 SHEET 1

DWG NO. SNPB-D-GRE-000-001

JOB NO.

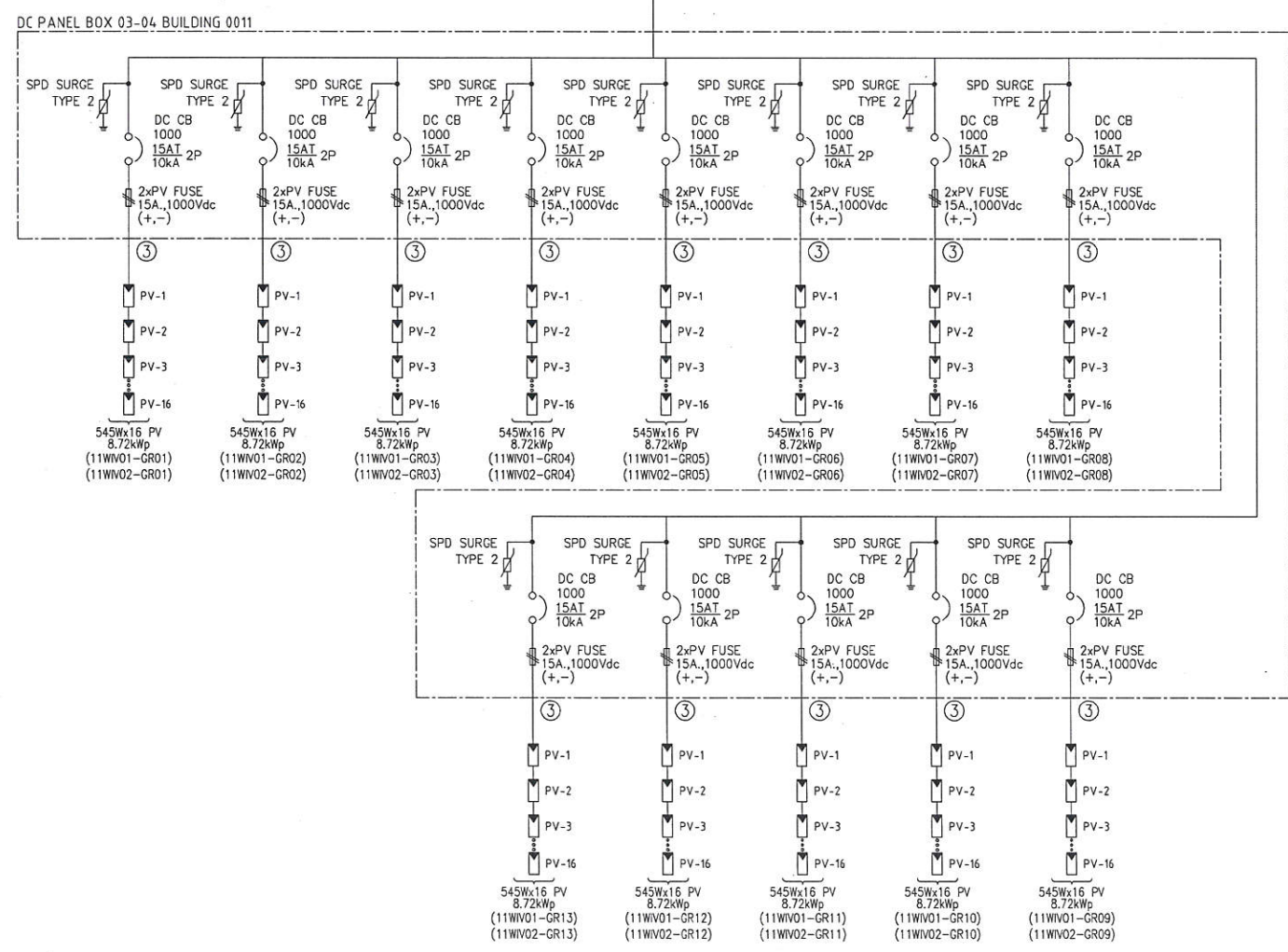
SHEET : 7/11 SIZE : A1 SCALE : NTS





DETAIL LP-02

- NOTES :
- 1. MAIN MCCB 3P 400AT/400AF (ADJUST TO 320AT) IS PROVIDED WITH CLOSING /TRIPPING COIL UNIT FOR RECEIVING SIGNAL FROM PROTECTION RELAYS AND TIMER.
 - 2. INVERTER NO.03 AND 04 USES A TOTAL OF 10 MPPT.
 - 3. MORE DETAIL OF MPPT CONNECTION SHALL REFER TO SNPB-D-GRE-WAQ-009 CABLE SCHEDULE.



- LEGEND DESCRIPTION
- SHT = SHUNT TRIP
 - DM = DIGITAL METER
 - EFR = EARTH FAULT RELAY

CABLE KEY NOTES :

CODE	
①	4Cx240 sq.mm. CV-FD(0.6/1.0kV) CABLE,1x25G sq.mm. IEC01 CABLE IN PERFORATED TRAY 200x100 mm. W/COVER
②	4Cx95 sq.mm. CV-FD(0.6/1.0kV), G1x16 sq.mm. IEC01 CABLE IN PERFORATED TRAY 200x100 mm. W/COVER
③	2x6 sq.mm. PV1-F CABLE IN PERFORATED TRAY 100x100 mm. W/COVER
④	2-10/6G IEC01 IN Ø3/4" FLEX
⑤	4C-10 sq.mm. CV CABLE, G1x6 sq.mm. IEC01 CABLE IN Ø1 1/4" IMC
⑥	2-10/6G IEC01 IN Ø1/2" FLEX
⑦	2 CORES 16AWG SHIELD OUTDOOR TYPE IN Ø1/2" FLEX
⑧	13(1x6 sq.mm. PV1-F CABLE) IN Ø1 1/4" FLEX (POSITIVE)
	13(1x6 sq.mm. PV1-F CABLE) IN Ø1 1/4" FLEX (NEGATIVE)

PROJECT NAME :
SAI NOI PV&BESS PROJECT

LOCATION :
EGAT Sai Noi Office at 81 Moo 11 Bangkrui-Sai Noi Road, Sai Noi Sub-district, Sai Noi District Nonthaburi Province, Thailand.

OWNER :

Electricity Generating Authority of Thailand

CONTRACTOR :

RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN

NOTE.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
6	FOR APPROVAL	24.12.2023	Phonabn	MarthaA	MarthaA
7	FOR CONSTRUCTION	11.01.2024	Phonabn	MarthaA	MarthaA
8	AS-BUILT	22.04.2024	Phonabn	MarthaA	MarthaA

DESCRIPTION :
LV SINGLE LINE DIAGRAM
81719 1.0011

DWG NO. SNPB-D-GRE-000-001

JOB NO.

SHEET : 9/11 SIZE : A1 SCALE : NTS

18SNPB-WAC01
(PV-ACDB-03 BUILDING 0018)

ALL CB IC > 30 kA
AT 415 VAC
IP41

3x SURGE ARRESTER
(TYPE 1)

HRC
3x FUSE 125A

3x FUSE 6A

MCCB
3P, 200AT
250AF

Adj 160A

4x CT
200/5A

3x CT
200/5A

MAIN BUS 250A WITH 100%N, 25%Z

F1

F2

F3

F4

F5

F6

MCCB.1
3P, 200AT
250AF

MCCB.2
3P, 50AT
100AF

MCCB.3
3P, 50AT
100AF

MCCB.5
2P, 50AT
100AF

INVERTER NO.05
(SEE DETAIL D)

BOOSTER PUMP
3kW, 3Ph

SPACE

SPACE

LP-03

SPACE

NOTES : 1. MAIN MCCB 3P 200AT/250AF (ADJUST TO 160AT) IS PROVIDED WITH CLOSING /TRIPPING COIL UNIT FOR RECEIVING SIGNAL FROM PROTECTION RELAYS AND TIMER.
2. INVERTER NO.05 USES A TOTAL OF 10 MPPT.
3. MORE DETAIL OF MPPT CONNECTION SHALL REFER TO SNPB-D-GRE-WAQ-009 CABLE SCHEDULE.

18SNPB-WAC01
(PV-ACDB-03 BUILDING 0018)

18SNPB-WIV01
(INVERTER No.05)

HUAWEI
100kW(100kVA) 3Ph 4W+E, 400V,144.34A 50Hz

AC SURGE PROTECTION DEVICE

18SNPB-WDC01
(DC PANEL BOX 05 BUILDING 0018)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR01)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR02)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR03)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR04)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR05)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR06)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR07)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR08)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR09)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR10)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR11)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR12)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR13)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR14)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR15)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR16)

SPD SURGE TYPE 2

DC CB
1000
15AT 2P
10kA

2xPV FUSE
15A,1000Vdc
(+,-)

PV-1

PV-2

PV-3

PV-18

545Wx18 PV
9.81kWp
(18WV01-GR17)

LEGEND DESCRIPTION

(SHT) = SHUNT TRIP

(DM) = DIGITAL METER

(EFR) = EARTH FAULT RELAY

12 STRINGS/216 PV PANELS (117.72kWp)

DETAIL D - INVERTER 100kW

18SNPB-WAC01
(PV-ACDB-03 BUILDING 0018)

18SNPB-WLP01
(LP-03)

Isc > 25kA.
IP 54

MCCB
2P,50AT/50AF

100A CU. BUSBAR 1Ph, 100%N, 25% GROUND

F1

F2

F3

F4

F5

F6

MCB
1P, 20AT
50AF

MCB
1P, 20AT
50AF

MCB
1P, 16AT
50AF

MCB
2P, 20AT
50AF

SPACE

SPACE

UPS FOR RTU PANEL
800VA 1Ph

DETAIL LP-03

CABLE KEY NOTES :

CODE	
①	4Cx70 sq.mm. CV-FD(0.6/1.0kV) CABLE,1x16G sq.mm. IEC01 CABLE IN WIREWAY 150x100 mm. W/COVER
②	4Cx95 sq.mm. CV-FD(0.6/1.0kV) CABLE,1x16G sq.mm. IEC01 CABLE IN WIREWAY 200x100 mm. W/COVER
③	2x6 sq.mm. PV1-F CABLE IN PERFORATED TRAY 100x100 mm. W/COVER
④	2-10/6G IEC01 IN Ø3/4" IMC
⑤	4C-10 sq.mm. CV CABLE, G1x6 sq.mm. IEC01 CABLE IN Ø32 mm. HDPE PIPE
⑥	2-10/6G IEC01 IN Ø1/2" FLEX
⑦	2 CORES 16AWG SHIELD OUTDOOR TYPE IN Ø1/2" EMT
⑧	12(1x6 sq.mm. PV1-F CABLE) IN Ø1 1/4" FLEX (POSITIVE)
	12(1x6 sq.mm. PV1-F CABLE) IN Ø1 1/4" FLEX (NEGATIVE)

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



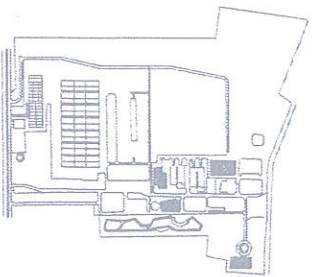
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Handwritten signature and notes in Thai script.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
6	FOR APPROVAL	21/08/2023	Phumadua	NorthDA	NorthDA
7	FOR CONSTRUCTION	21/08/2023	Phumadua	NorthDA	NorthDA
8	AS-BUILT	21/08/2023	Phumadua	NorthDA	NorthDA

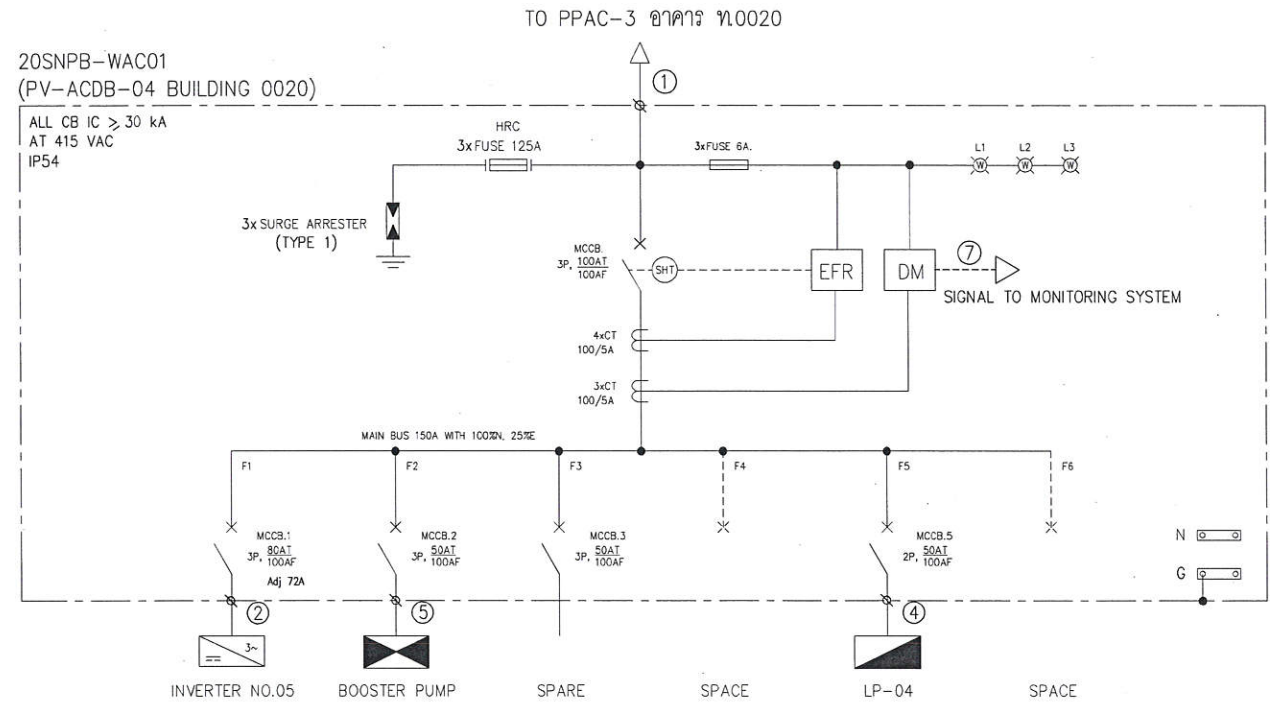
DESCRIPTION :

LV SINGLE LINE DIAGRAM
01/01/2018

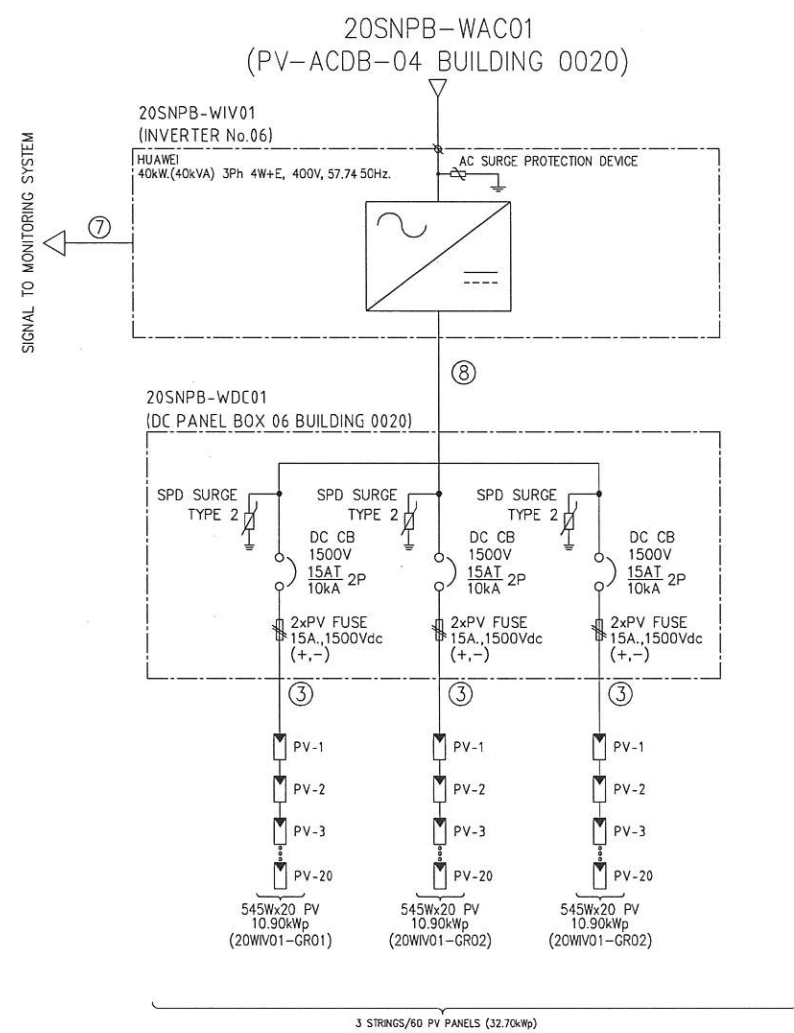
DWG NO. SNPB-D-GRE-000-001

JOB NO.

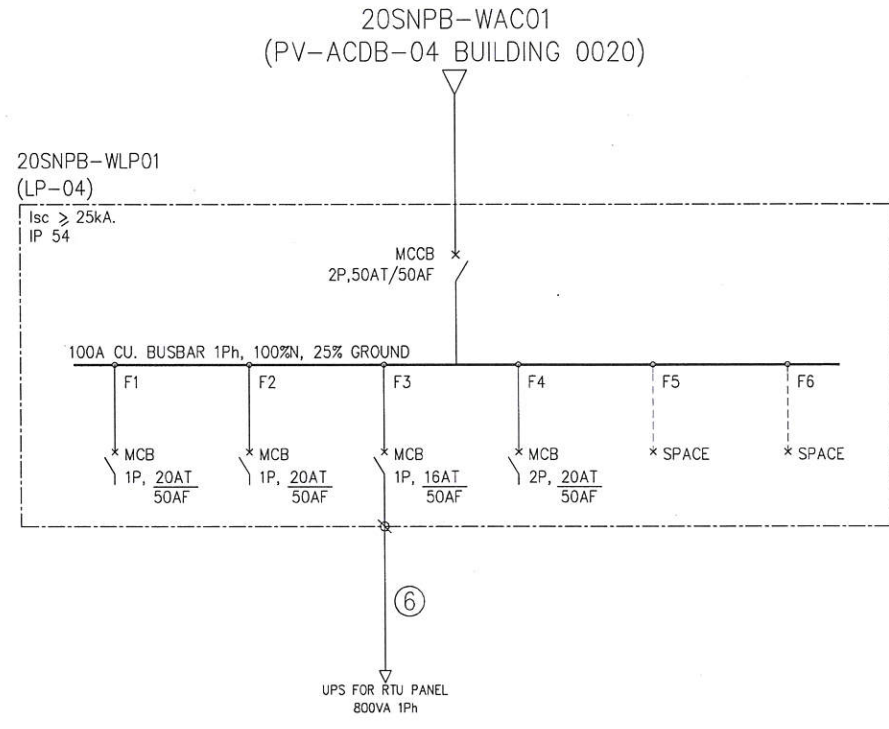
SHEET : 10/11 SIZE : A1 SCALE : NTS



1. MAIN MCCB 3P 100AT/100AF IS PROVIDED WITH CLOSING /TRIPPING COIL UNIT FOR RECEIVING SIGNAL FROM PROTECTION RELAYS AND TIMER.
2. INVERTER NO.06 USES A TOTAL OF 3 MPPT.
3. MORE DETAIL OF MPPT CONNECTION SHALL REFER TO SNPB-D-GRE-WAQ-009 CABLE SCHEDULE.



DETAIL E - INVERTER 40kW



DETAIL LP-04

CABLE KEY NOTES :

CODE	
①	4Cx25 sq.mm. CV-FD(0.6/1.0kV) CABLE,1x10G sq.mm. IEC01 CABLE IN Ø2" FLEX
②	4Cx25 sq.mm. CV-FD(0.6/1.0kV) CABLE,1x10G sq.mm. IEC01 CABLE IN SPIRAL CABLE WRAP
③	2x6 sq.mm. PV1-F CABLE IN PERFORATED TRAY 100x100 mm. W/COVER
④	2-10/6G IEC01 IN Ø3/4" FLEX
⑤	4C-10 sq.mm. CV CABLE, G1x6 sq.mm. IEC01 CABLE IN Ø1 1/4" IMC
⑥	2-10/6G IEC01 IN Ø1/2" FLEX
⑦	2 CORES 16AWG SHIELD OUTDOOR TYPE IN Ø1/2" FLEX
⑧	3(2x6 sq.mm. PV1-F CABLE) IN Ø1 1/4" FLEX

- LEGEND DESCRIPTION
- SHT = SHUNT TRIP
 - DM = DIGITAL METER
 - EFR = EARTH FAULT RELAY

PROJECT NAME :
SAI NOI PV&BESS PROJECT

LOCATION :
EGAT Sai Noi Office at 81 Moo 11 Bangkrui-Sai Noi Road, Sai Noi Sub-district, Sai Noi District Nonthaburi Province, Thailand.

OWNER :
Electricity Generating Authority of Thailand

CONTRACTOR :
RATCH GROUP
RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN

NOTE.

REVISION

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
6	FOR APPROVAL	21/08/2023	Phumadon	NorthA	NorthA
7	FOR CONSTRUCTION	21/08/2023	Phumadon	NorthA	NorthA
8	AS-BUILT	21/08/2023	Phumadon	NorthA	NorthA

DESCRIPTION :
LV SINGLE LINE DIAGRAM
๑๗๑๓ ๗๐๐๒๐

DWG NO. SNPB-D-GRE-000-001

JOB NO.

SHEET : 11/11 SIZE : A1 SCALE : NTS

ใบอนุญาตประกอบวิชาชีพวิศวกรรมควบคุม
Thai Professional Engineering License

เลขประจำตัวประชาชน (ID) 3 1201 00425 80 5

ชื่อและนามสกุล
Title/Name Surname **นายมลุต เอี่ยมสำอางค์**
Mr. Monthol Aemsam-ang

เลขทะเบียน วฟก.1248 เลขที่สมาชิกสามัญ 34954
License No. Member No.

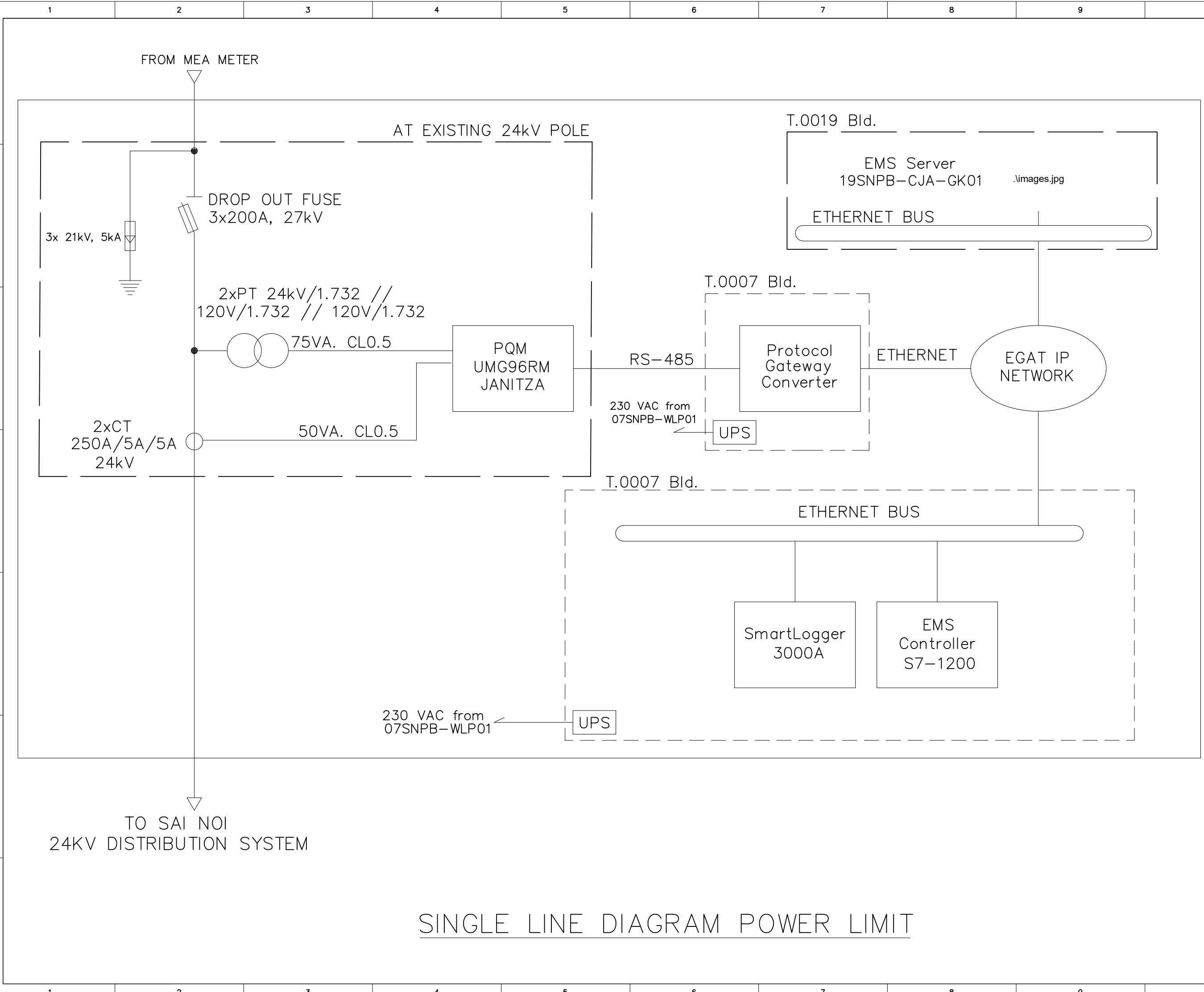
ระดับ วุฒิศวกร สาขา ไฟฟ้ากำลังไฟฟ้ากำลัง
Level Senior Professional Discipline Electrical-Eng. (Power)

วันอนุญาต 30 พ.ค. 2562 วันหมดอายุ 29 พ.ค. 2567
Date of Issue 30 May. 2019 Date of Expiry 29 May. 2024

(ลายมือชื่อในใบอนุญาต) (นายมลุต เอี่ยมสำอางค์)
ลายมือชื่อในใบอนุญาต (Signature) นายมลุต เอี่ยมสำอางค์
President

เพื่อใช้ในการรับรองแบบ Single Line Diagram
โครงการ Sainoi PV&BESS Project

5	23/06/2023	As-Built					
4	21/06/2023	Final for Construction					
3	29/03/2023	Issued for Approval					
Rev.	Date	Description					
Owner : ELECTRICITY GENERATING AUTHORITY OF THAILAND							
Contractor : RATCH Group Public Company Limited							
ARCHITECT : -		SUBMITTED		SAI NOI PV AND BESS PROJECT			
ENGINEER : -							
ENGINEER : -		CONCURRED		Drawing name Zero Export Controller Diagram			
DRAWN : -							
CHECKED : -				REPLACING DWG. NO. -	DWG. NO. SNPB-D-GRE- 000-004	REV NO.	SHEET NO.
APPROVED				DATE		5	1/2



PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81 Moo 11 Bangkrui-Sai Noi Road, Sai Noi Sub-district, Sai Noi District Nonthaburi Province, Thailand.

OWNER :



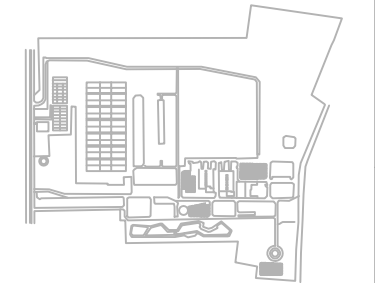
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH Group Public Company Limited

KEYPLAN



NOTE.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
3	FOR APPROVAL	29/03/2023	PREWADOLUX	MONTHOLA	MONTHOLA
4	FOR CONSTRUCTION	21/08/2023	PREWADOLUX	MONTHOLA	MONTHOLA
5	AS-BUILT	23/08/2023	PREWADOLUX	MONTHOLA	MONTHOLA

DESCRIPTION :

Zero Export Controller Diagram

DWG NO. SNPB-D-GRE-000-004

JOB NO.-

SHEET : 1/1

SIZE : A3

SCALE : NTS

SINGLE LINE DIAGRAM POWER LIMIT

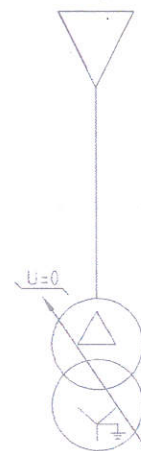
4	23/06/2023	As-Built				
3	23/06/2023	Final for Construction				
2	21/06/2023	Issued for Approval				
1	09/06/2023	Issued for Approval				
0	12/04/2022	Issued for Approval				
Rev.	Date	Description				
Owner : ELECTRICITY GENERATING AUTHORITY OF THAILAND						
Contractor : RATCH Group Public Company Limited						
ARCHITECT : -		SUBMITTED	SAI NOI PV AND BESS PROJECT			
ENGINEER : -						
ENGINEER : -		CONCURRED	Drawing name Outline Drawing for BESS and PCS			
DRAWN : -						
CHECKED : -						
APPROVED		DATE.....	REPLACING DWG. NO. -	DWG. NO. SNPB-D-GRE- 000-005	REV NO. 4	SHEET NO. 1/2

4	23/06/2023	As-Built					
3	22/06/2023	Final for Construction					
2	21/06/2023	Issued for Approval					
1	13/06/2023	Issued for Approval					
0	05/03/2022	Issued for Approval					
Rev.	Date	Description					
Owner : ELECTRICITY GENERATING AUTHORITY OF THAILAND							
Contractor : RATCH Group Public Company Limited							
ARCHITECT : -		SUBMITTED		SAI NOI PV AND BESS PROJECT			
ENGINEER : -							
ENGINEER : -		CONCURRED		Drawing name AC Single Line Diagram for BESS and PCS			
DRAWN : -							
CHECKED : -		APPROVED		REPLACING DWG. NO. -	DWG. NO. SNPB-D-GRE- 000-006	REV NO. 4	SHEET NO. 1/3
DATE							

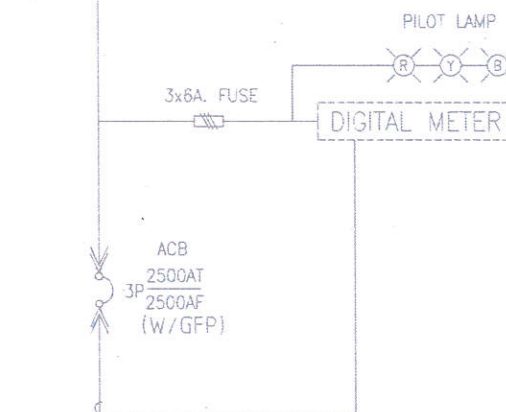

 น.ก. บัณฑิต
 ๗๓. ๑๒๔๘

SUS

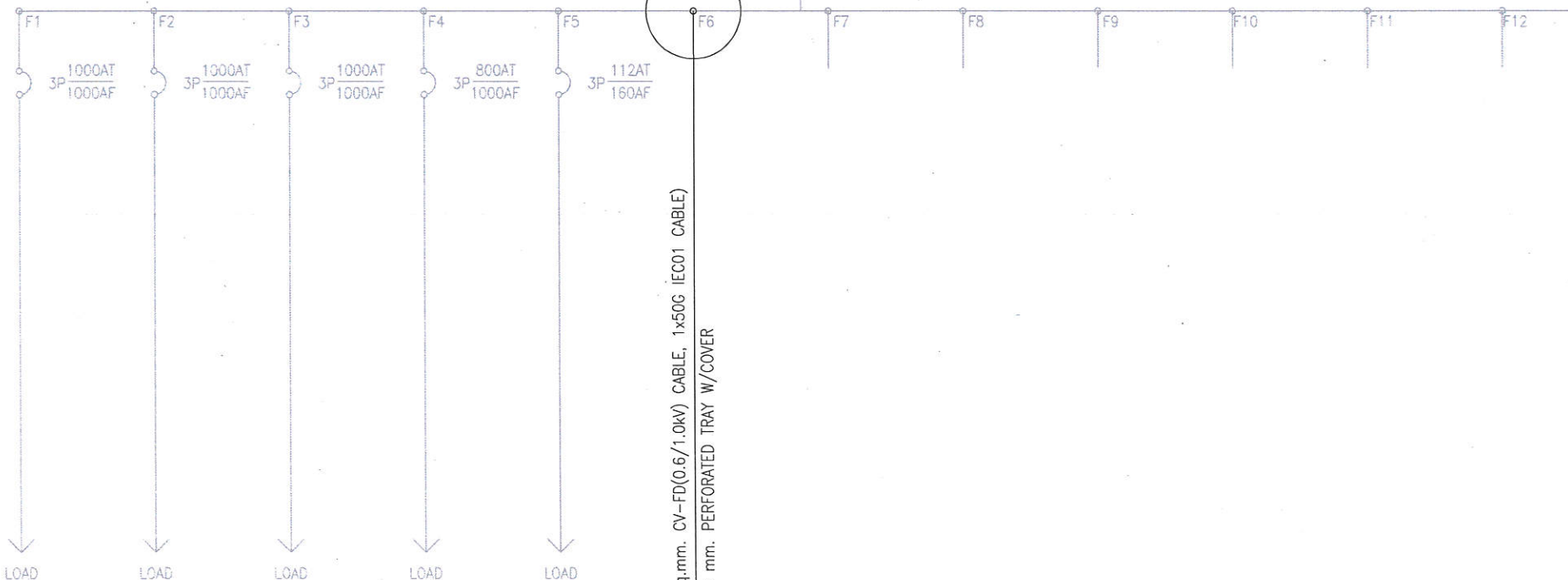
MAIN IN-COMING
SUPPLY BY EGAT
3 PHASE 4 WIRES 416/240 Vac., 50 Hz.



TRANSFORMER
OIL-IMMERSED TYPE
1500 kVA (ONAN) Dyn11
24kV PRI./416Y/2245V SEC., 50Hz



MODIFY EXISTING BUSBAR



2(4Cx185 sq.mm. CV-FD(0.6/1.0KV) CABLE, 1x50G IEC01 CABLE)
IN 200x100 mm. PERFORATED TRAY W/COVER
BESS
300kW

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



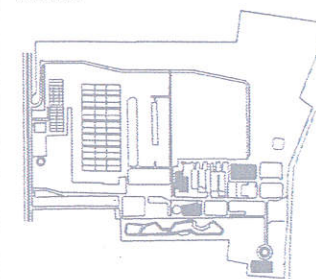
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Signature
วันที่ ๒๐ ธันวาคม ๒๕๖๓
พ. ๑๒๔๘

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
0	FOR APPROVAL	01/06/2563	SARATUTP	SARATUTP	PLDN
1	FOR APPROVAL	13/06/2563	Phumadok	Phumadok	MontolA
2	FOR APPROVAL	21/06/2563	Phumadok	Phumadok	MontolA
3	FOR CONSTRUCTION	21/06/2563	Phumadok	Phumadok	MontolA
4	AS-BUILT	01/06/2563	Phumadok	Phumadok	MontolA

DESCRIPTION :

AC Single Line Diagram for BESS and PCS

DWG NO. SNPB-D-GRE-000-006

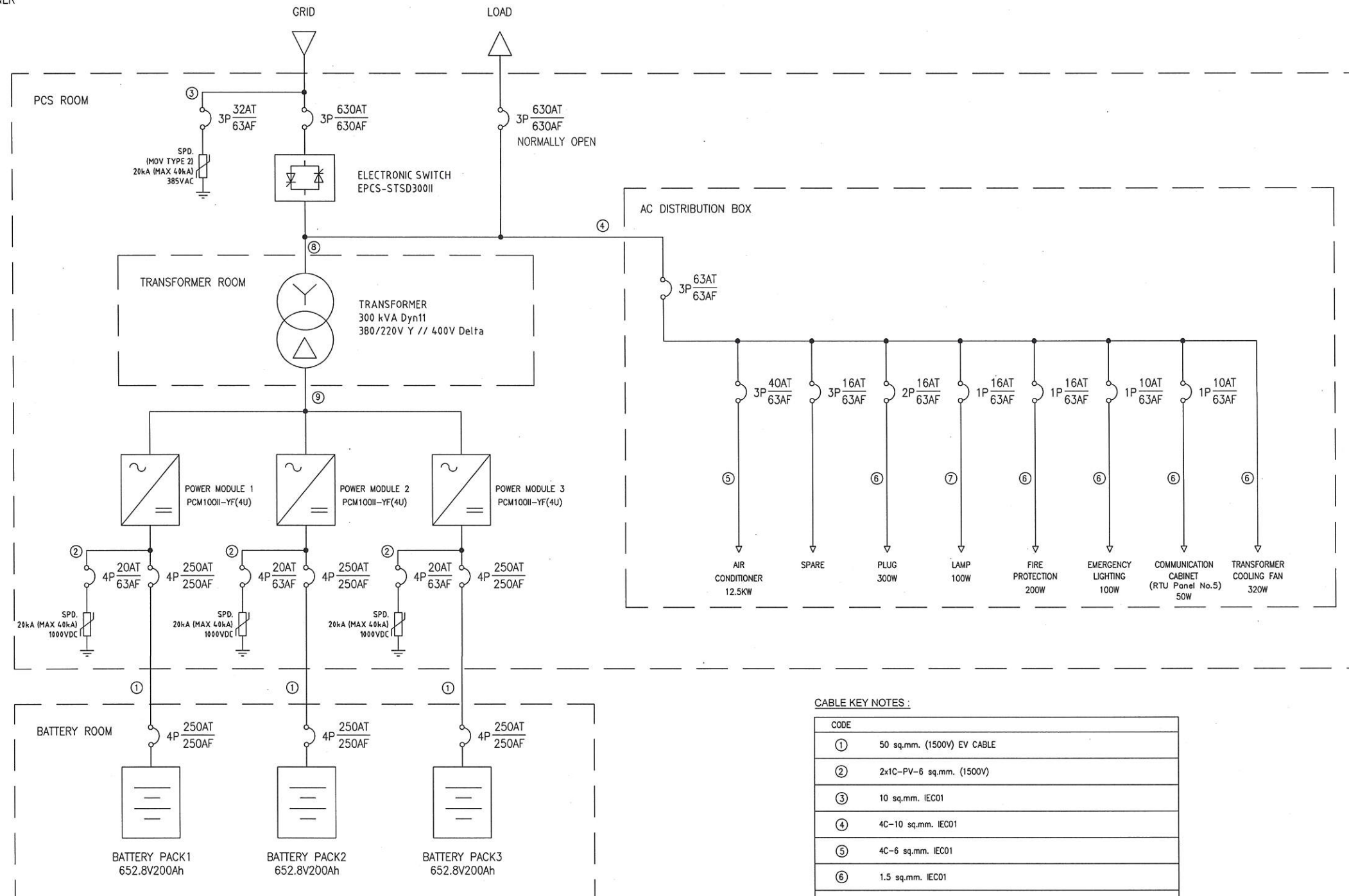
JOB NO.

SHEET : 1

SIZE : A3

SCALE : NTS

BESS CONTAINER



CABLE KEY NOTES :

CODE	
①	50 sq.mm. (1500V) EV CABLE
②	2x1C-PV-6 sq.mm. (1500V)
③	10 sq.mm. IEC01
④	4C-10 sq.mm. IEC01
⑤	4C-6 sq.mm. IEC01
⑥	1.5 sq.mm. IEC01
⑦	1 sq.mm. IEC01
⑧	2(4x1C-95 sq.mm. 60227 IEC 02(RV) (450/750V) CABLE)
⑨	2(3x1C-95 sq.mm. 60227 IEC 02(RV) (450/750V) CABLE)

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkrui-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



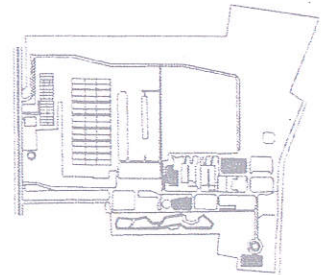
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Handwritten signature and notes:
นาย อดิศักดิ์ ใจดี
วันที่ 12/4/25

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
0	FOR APPROVAL	21/02/2023	SARATUTP	SARATUTP	PLDEN
1	FOR APPROVAL	12/04/2023	PhumadokA	PhumadokA	MeritA
2	FOR APPROVAL	11/06/2023	PhumadokA	PhumadokA	MeritA
3	FOR CONSTRUCTION	21/08/2023	PhumadokA	PhumadokA	MeritA
4	AS-BUILT	21/08/2023	PhumadokA	PhumadokA	MeritA

DESCRIPTION :

AC Single Line Diagram for BESS and PCS

DWG NO. SNPB-D-GRE-000-006

JOB NO.

SHEET : 2 SIZE : A3 SCALE : NTS

ใบอนุญาตประกอบวิชาชีพวิศวกรรมควบคุม
Thai Professional Engineering License

เลขประจำตัวประชาชน (ID) 3 1201 00425 80 5

ชื่อตัวและชื่อสกุล
Title/Name Surname นายมณฑล เอี่ยมสำอางค์
Mr. Monthol Aemsam-ang

เลขทะเบียน วพก.1248 เลขที่สมาชิกสามัญ 35494
License No. Member No.

ระดับ วิศวกร สาขา ไฟฟ้า งานไฟฟ้ากำลัง
Level Senior Professional Discipline Electrical Eng. (Power)

วันอนุญาต 30 พ.ค. 2562 วันหมดอายุ 29 พ.ค. 2567
Date of Issue 30 May. 2019 Date of Expiry 29 May. 2024

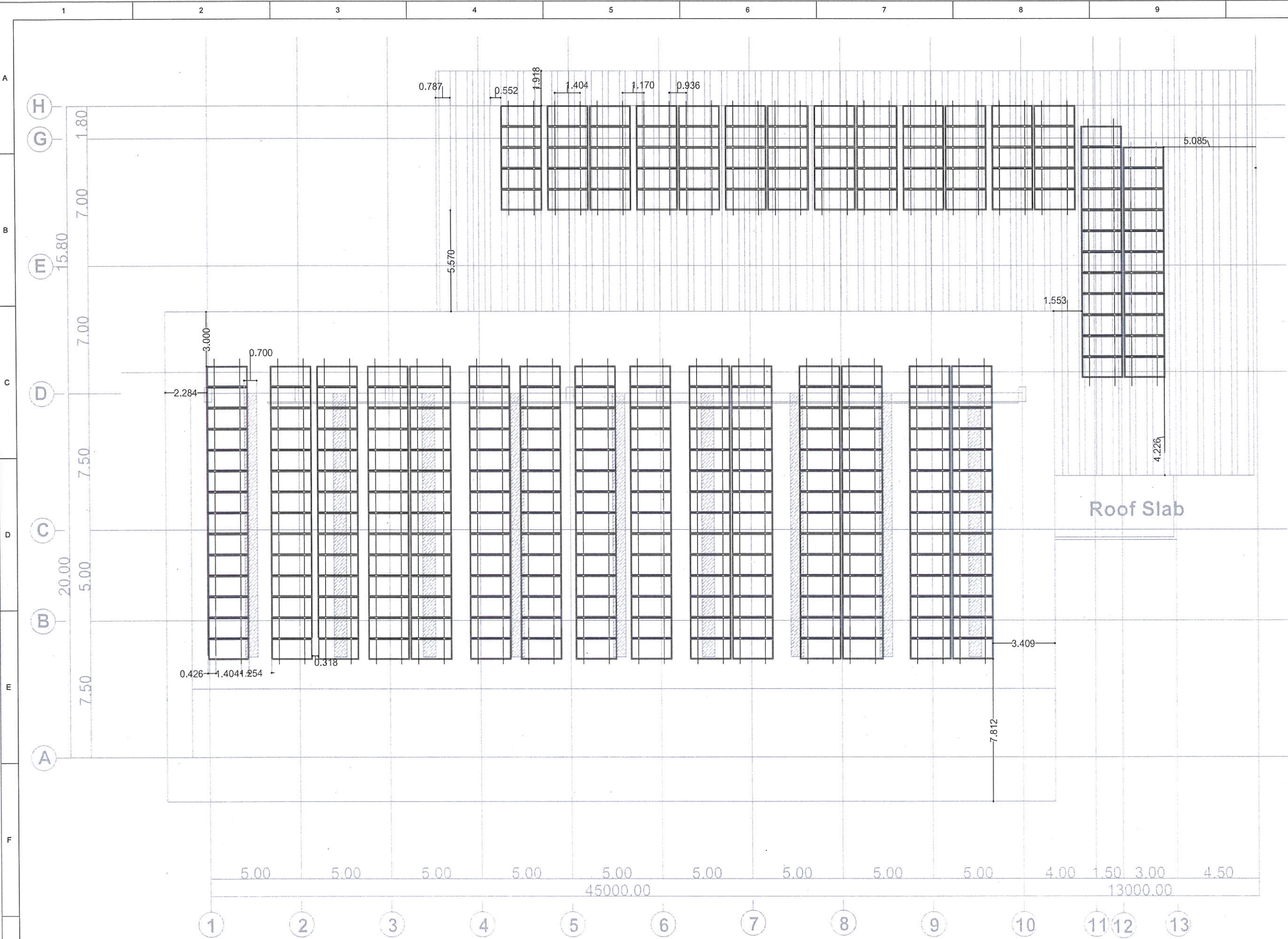
(ลายเซ็น)
ลายมือชื่อผู้ได้รับใบอนุญาต (Signature)

(นายมณฑล เอี่ยมสำอางค์)
นายกสภาวิศวกร
President

เพื่อใช้ในการยื่นการรับรองแบบ AC Single Line Diagram for BESS and PCS
โครงการ Sainoi PV&BESS Project

7	23/06/2023	As-Built				
6	21/06/2023	Final for Construction				
5	24/04/2023	Issued for Approval				
Rev.	Date	Description				
Owner : ELECTRICITY GENERATING AUTHORITY OF THAILAND						
Contractor : RATCH Group Public Company Limited						
ARCHITECT : -		SUBMITTED	SAI NOI PV AND BESS PROJECT			
ENGINEER : -						
ENGINEER : -		CONCURRED	Drawing name PV Layout			
DRAWN : -						
CHECKED : -						
APPROVED			REPLACING DWG. NO. -	DWG. NO. SNPB-D-GRE- WPA-001	REV NO. 7	SHEET NO. 1/9
.....		DATE				

Order L. Jinn
นางสาว ใจเย็นใจ
รพ. 1248



INV.NO.	NUMBER OF INV.	PV/STRING	PV/INV.	TOTAL PV.	TOTAL W.
01	1	(14X13)	182	182	99,190
02	1	(15x4), (14x4)	116	116	63,220
TOTAL	2	298	-	298	162,410

PV LAYOUT ๑๓๓๓ ๓.๐๐๐๗

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



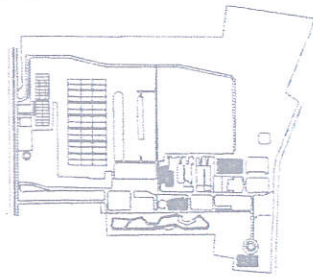
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE

Handwritten signature and notes in Thai script.
วันที่ 12/4/63

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
5	FOR APPROVAL	14/03/2023	Phumchonk	NorththA	NorththA
6	FOR CONSTRUCTION	21/04/2023	Phumchonk	NorththA	NorththA
7	AS-BUILT	21/04/2023	Phumchonk	NorththA	NorththA

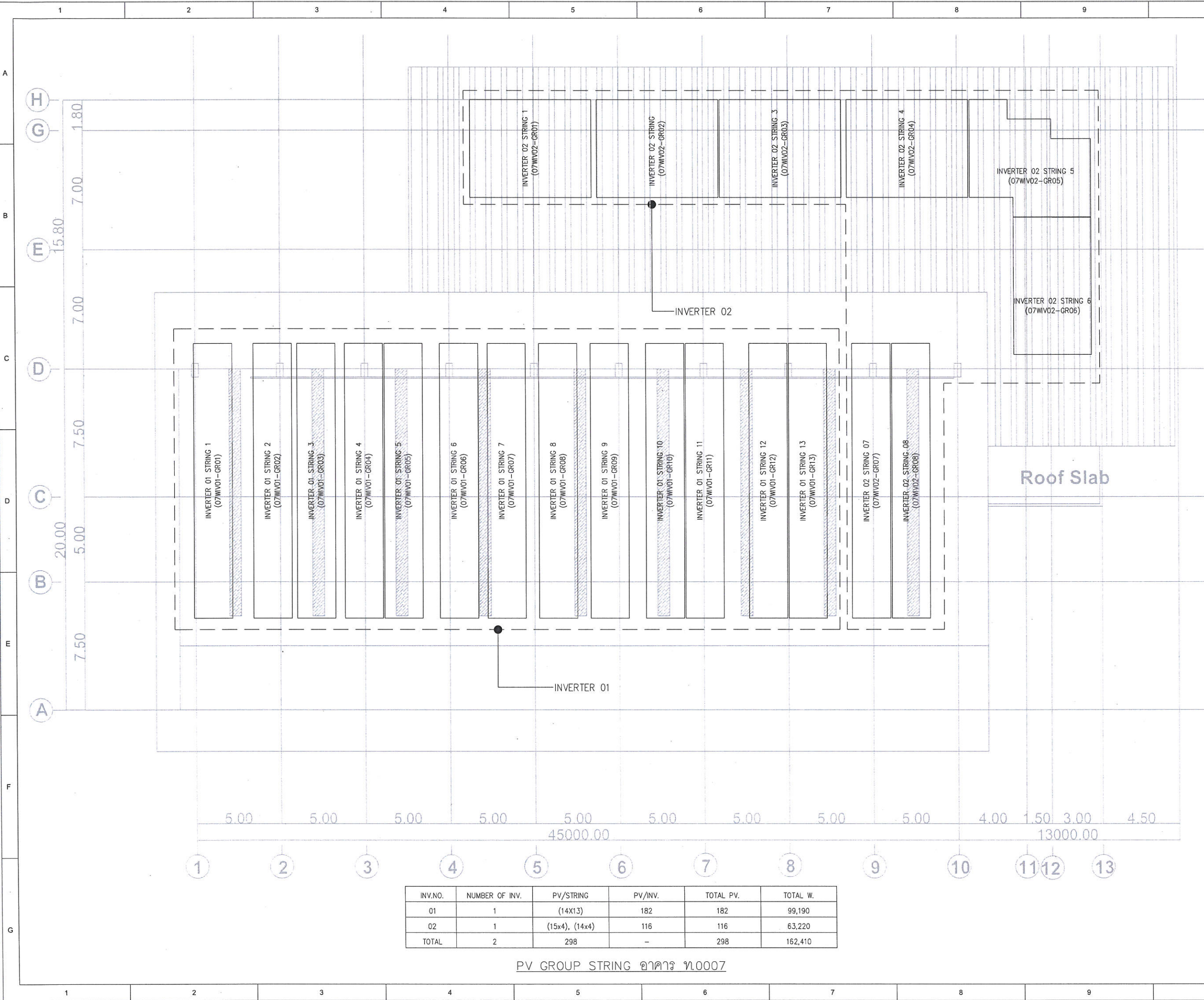
DESCRIPTION :

PV LAYOUT
๑๓๓๓ ๓.๐๐๐๗

DWG NO. SNPB-D-GRE-WPA-001

JOB NO.

SHEET : 1/8 SIZE : A1 SCALE : 1:100



INV.NO.	NUMBER OF INV.	PV/STRING	PV/INV.	TOTAL PV.	TOTAL W.
01	1	(14X13)	182	182	99,190
02	1	(15x4), (14x4)	116	116	63,220
TOTAL	2	298	-	298	162,410

PV GROUP STRING ๑๙๙๖ ๗.๐๐๐๗

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



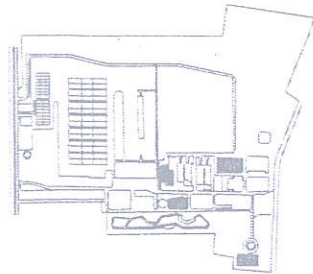
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Signature
นางสาว อธิษฐานพร
ทพ. 1248

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
5	FOR APPROVAL	11/10/2020	Phumthong	NorthA	NorthA
6	FOR CONSTRUCTION	11/10/2020	Phumthong	NorthA	NorthA
7	AS-BUILT	21/10/2020	Phumthong	NorthA	NorthA

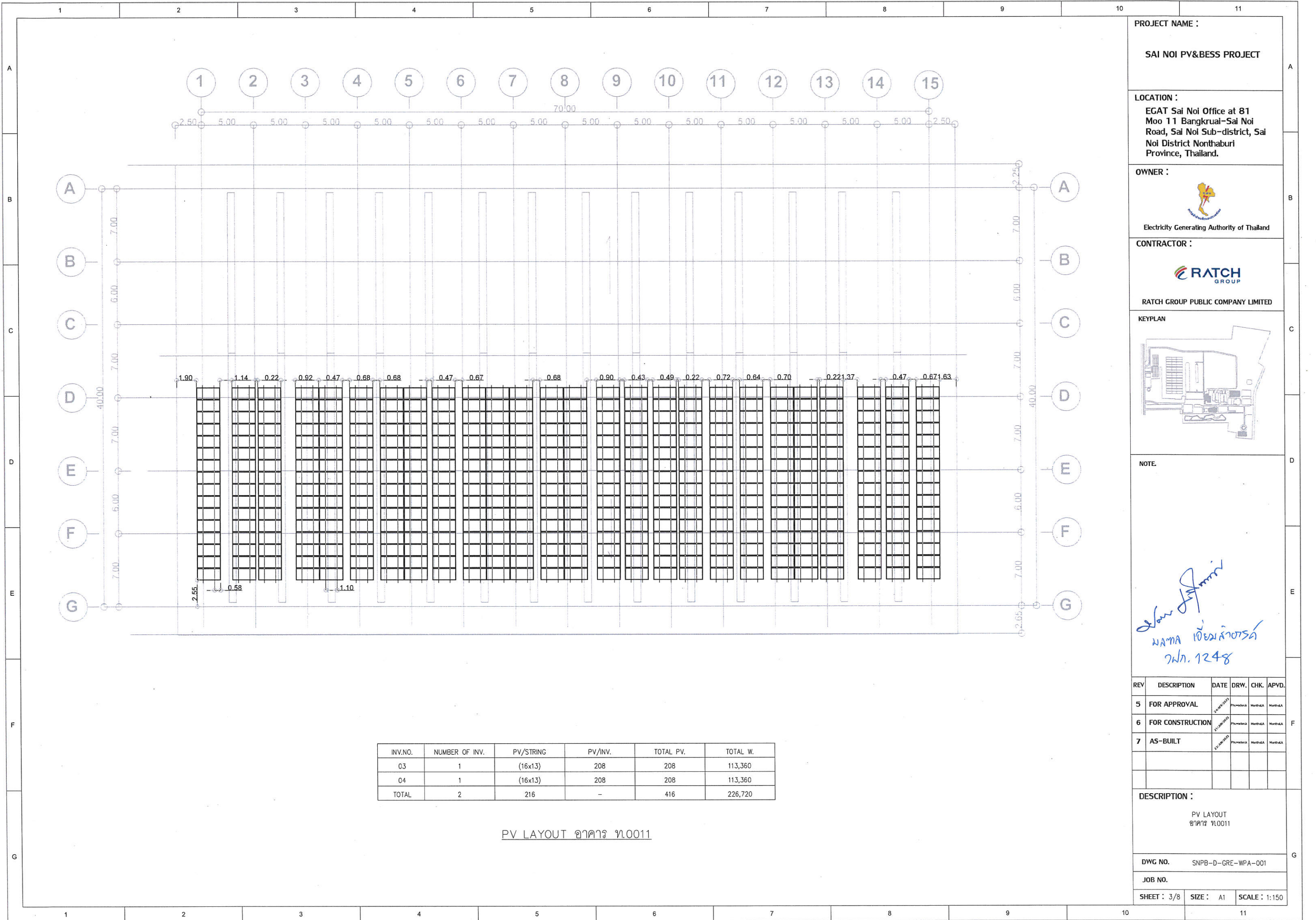
DESCRIPTION :

PV GROUP STRING
๑๙๙๖ ๗.๐๐๐๗

DWG NO. SNP8-D-GRE-WPA-001

JOB NO.

SHEET : 2/8 SIZE : A1 SCALE : 1:100



PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



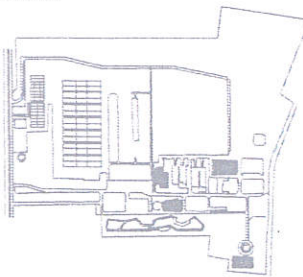
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Signature
นาย อดิศักดิ์
ท. 1248

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
5	FOR APPROVAL	24/04/2020	Phonadock	MartholA	MartholA
6	FOR CONSTRUCTION	21/06/2020	Phonadock	MartholA	MartholA
7	AS-BUILT	23/06/2020	Phonadock	MartholA	MartholA

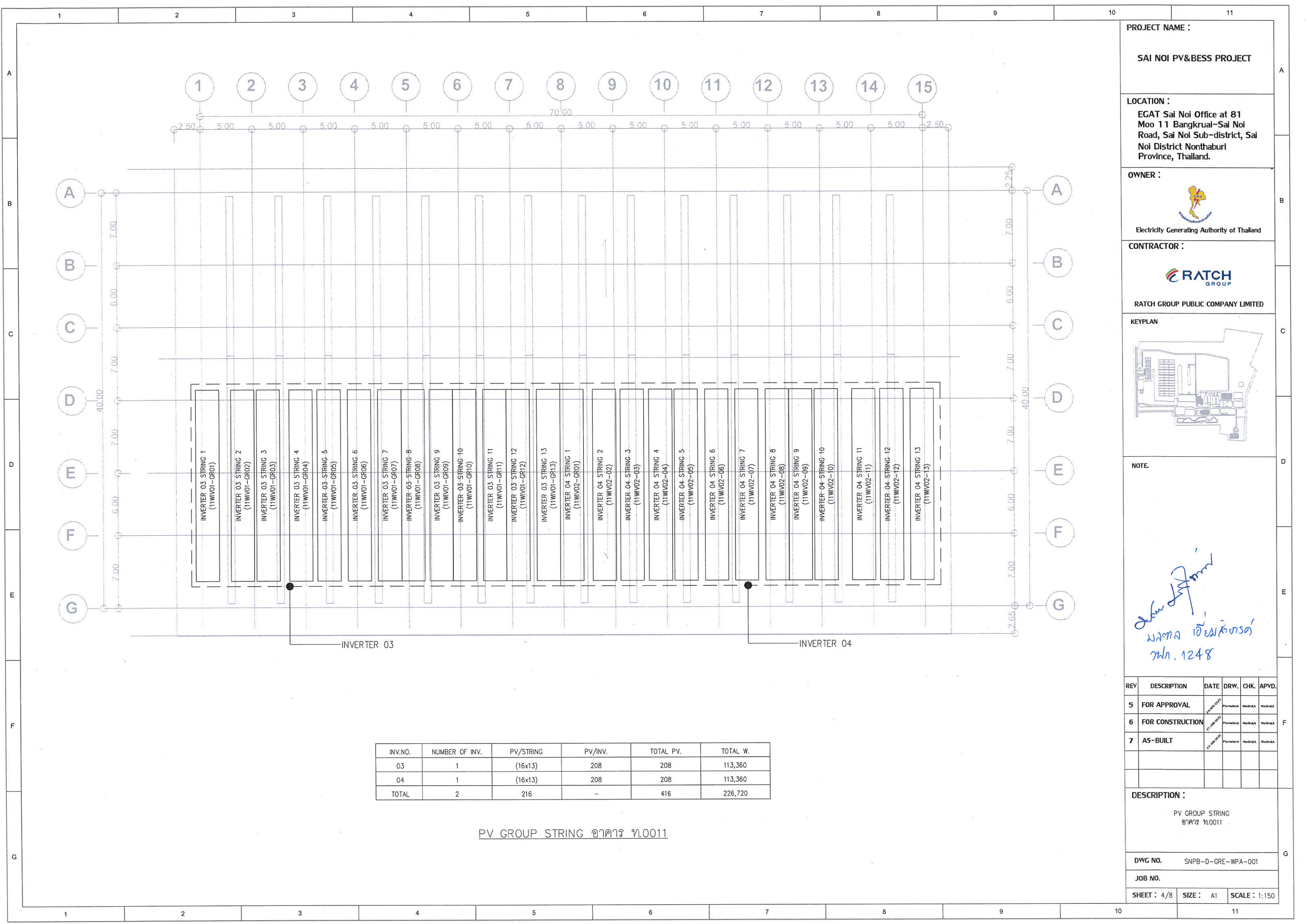
DESCRIPTION :

PV LAYOUT
๑๓๓๖ ๓.0011

DWG NO. SNPB-D-GRE-WPA-001

JOB NO.

SHEET : 3/8 SIZE : A1 SCALE : 1:150



PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkrual-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



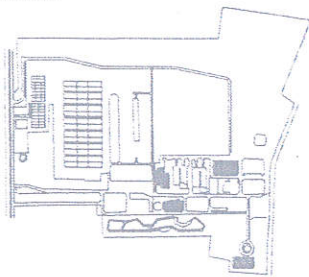
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Handwritten signature and note:
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พ.ร. 1248

INV.NO.	NUMBER OF INV.	PV/STRING	PV/INV.	TOTAL PV.	TOTAL W.
03	1	(16x13)	208	208	113,360
04	1	(16x13)	208	208	113,360
TOTAL	2	216	-	416	226,720

PV GROUP STRING ๑๓๓๖ ๗.0011

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
5	FOR APPROVAL	๒๕/๐๖/๒๕๖๒	Phomdara	Marthida	Marthida
6	FOR CONSTRUCTION	๒๖/๐๖/๒๕๖๒	Phomdara	Marthida	Marthida
7	AS-BUILT	๒๖/๐๖/๒๕๖๒	Phomdara	Marthida	Marthida

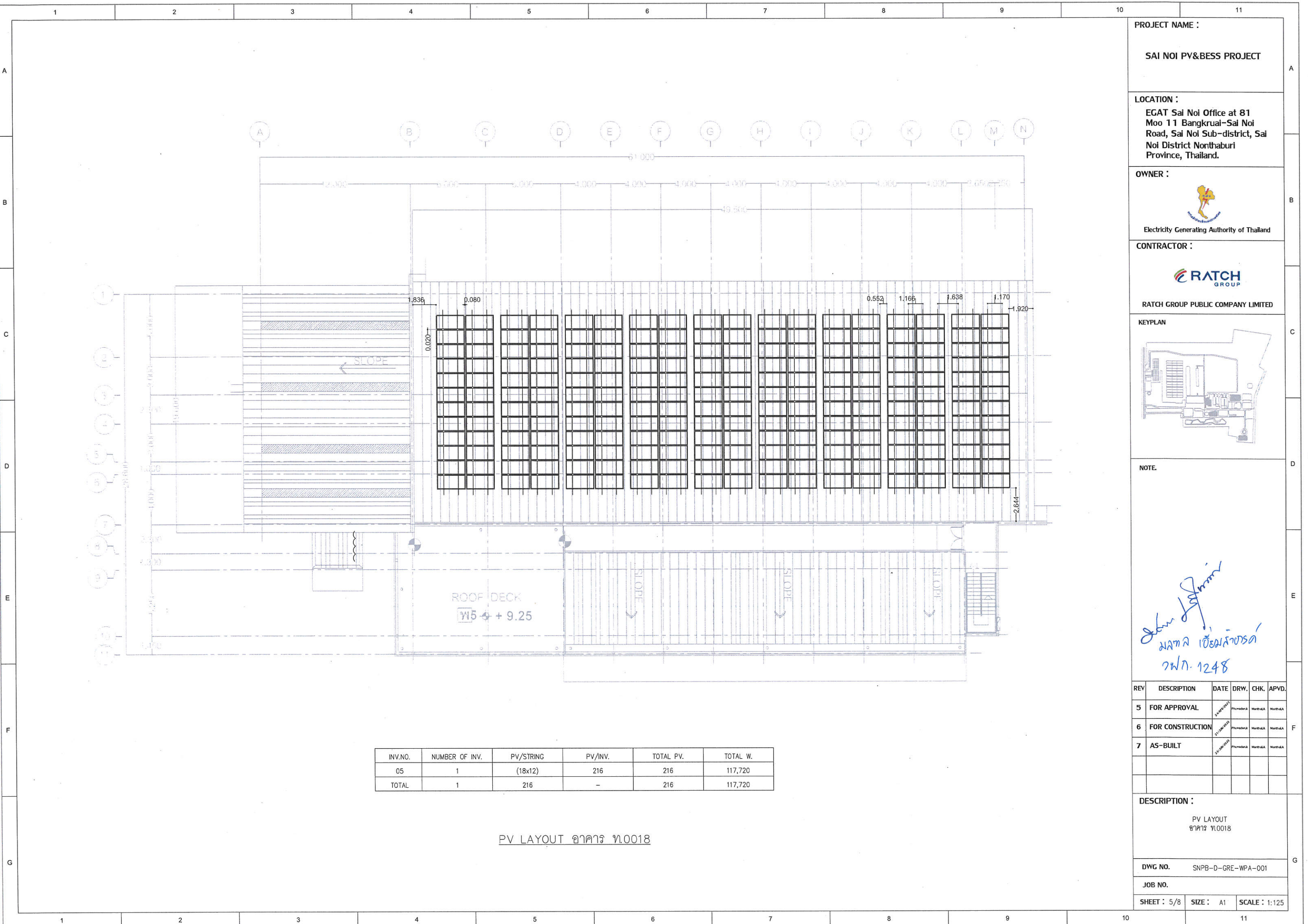
DESCRIPTION :

PV GROUP STRING
๑๓๓๖ ๗.0011

DWG NO. SNPB-D-GRE-WPA-001

JOB NO.

SHEET : 4/8 SIZE : A1 SCALE : 1:150



INV.NO.	NUMBER OF INV.	PV/STRING	PV/INV.	TOTAL PV.	TOTAL W.
05	1	(18x12)	216	216	117,720
TOTAL	1	216	-	216	117,720

PV LAYOUT ๑๓๓๓ ๓.๐๐18

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



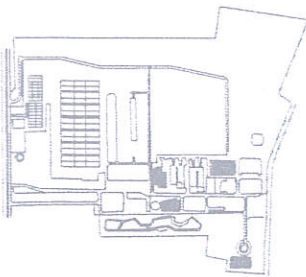
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

John J. Jinn
นาย วัฒนวิทย์
รพ. 1248

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
5	FOR APPROVAL	11/10/2018	Pumchonk	NorthA	NorthA
6	FOR CONSTRUCTION	11/10/2018	Pumchonk	NorthA	NorthA
7	AS-BUILT	11/10/2018	Pumchonk	NorthA	NorthA

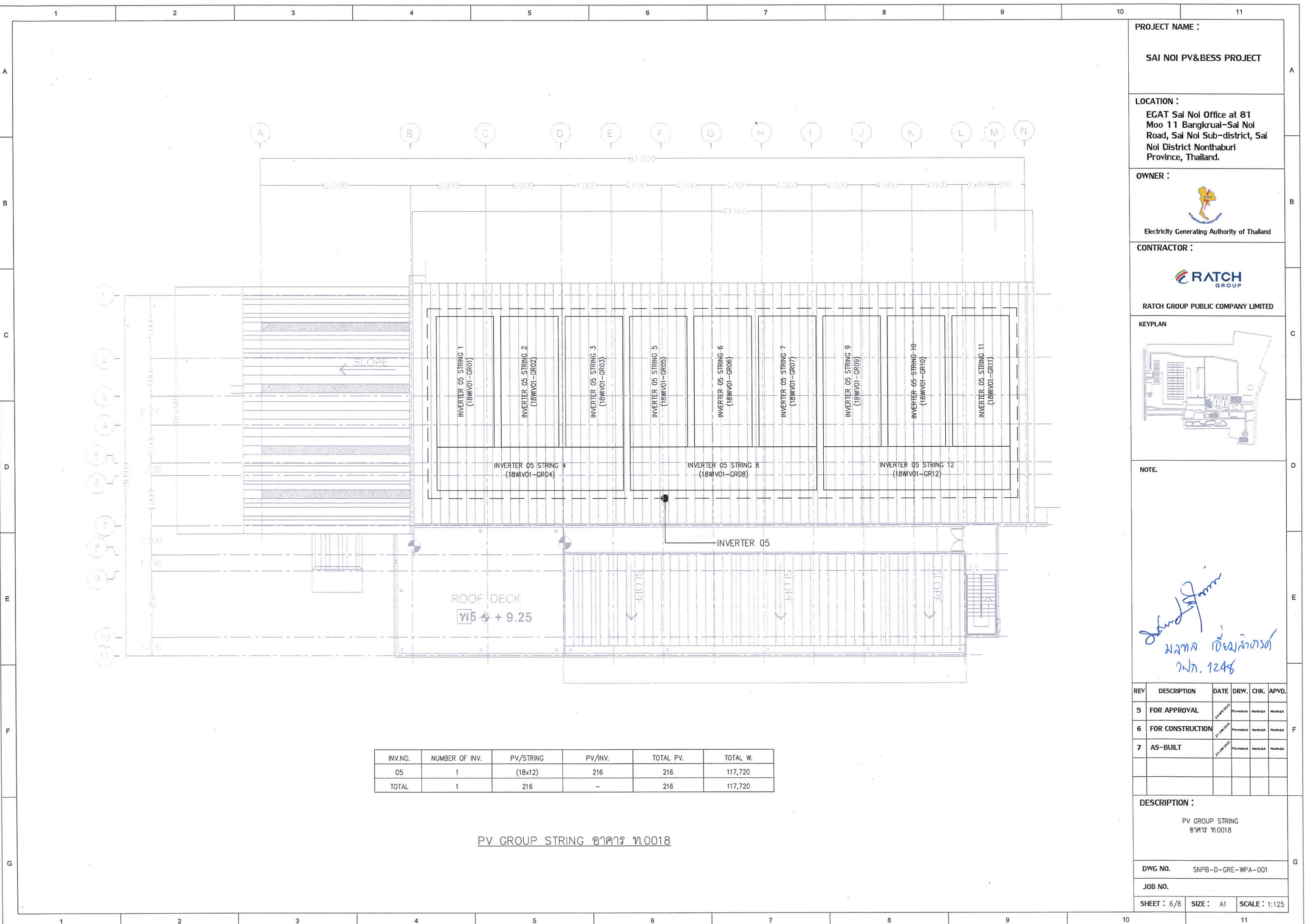
DESCRIPTION :

PV LAYOUT
๑๓๓๓ ๓.๐๐18

DWG NO. SNPB-D-GRE-WPA-001

JOB NO.

SHEET : 5/8 SIZE : A1 SCALE : 1:125



INV.NO.	NUMBER OF INV.	PV/STRING	PV/INV.	TOTAL PV.	TOTAL W.
05	1	(18x12)	216	216	117,720
TOTAL	1	216	-	216	117,720

PV GROUP STRING ตาราง ท.0018

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkrui-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



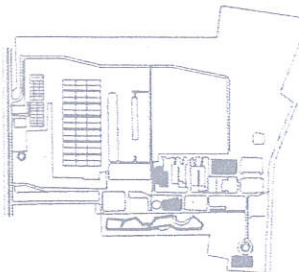
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE

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นางสาว อธิชา วัฒนศิริ
ท. 1248

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
5	FOR APPROVAL	24/07/2023	Phumthak	NorthA	NorthA
6	FOR CONSTRUCTION	11/08/2023	Phumthak	NorthA	NorthA
7	AS-BUILT	23/08/2023	Phumthak	NorthA	NorthA

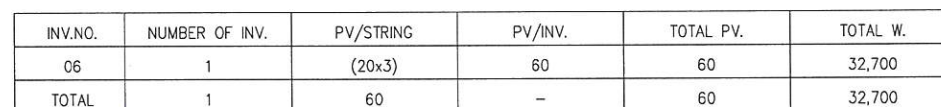
DESCRIPTION :

PV GROUP STRING
ตาราง ท.0018

DWG NO. SNPB-D-GRE-WPA-001

JOB NO.

SHEET : 6/8 SIZE : A1 SCALE : 1:125



PV GROUP STRING อาคาร ท.0020

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkruai-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



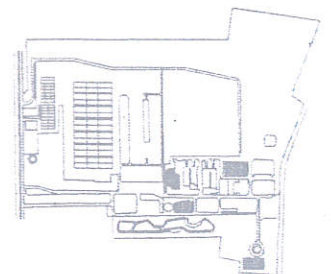
Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEYPLAN



NOTE.

Handwritten signature and date: ၁၂/၈. ၁၂၄၆

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
5	FOR APPROVAL	21-APR-2023	Plumadeira	NorthA	NorthA
6	FOR CONSTRUCTION	21-APR-2023	Plumadeira	NorthA	NorthA
7	AS-BUILT	21-APR-2023	Plumadeira	NorthA	NorthA

DESCRIPTION :	
---------------	--

PV GROUP STRING
๒1๗17 ๗0020

DWG BY.	DGN BY.	CHK BY.	APP BY.
KWANPONG P.	KYANPONG P.	SURASAK R.	SURASAK R.
DWG NO. SNPB-D-GRE-WPA-001			
JOB NO.			
SHEET : 8/8	SIZE : A1	SCALE : 1:125	

ใบอนุญาตประกอบวิชาชีพวิศวกรรมควบคุม
Thai Professional Engineering License

เลขประจำตัวประชาชน (ID) 3 1201 00425 80 5

ชื่อตัวและชื่อสกุล
Title/Name
Surname **นายมงคล เอี่ยมสำอางค์**
Mr. Monthol Aemsam-ang

เลขทะเบียน วฟก.1248 เลขที่สมาชิกสามัญ 35434
License No. Member No.

ระดับ วุฒิวิศวกร สาขา ไฟฟ้า พลังไฟฟ้า
Level Senior Professional Discipline Electrical Eng. (Power)

วันอนุญาต 30 พ.ค. 2562 วันหมดอายุ 29 พ.ค. 2567
Date of Issue 30 May. 2019 Date of Expiry 29 May. 2024

(นายมงคล เอี่ยมสำอางค์)
ลายมือชื่อผู้ได้รับใบอนุญาต (Signature) (นายสุวิทย์ สุวรรณศรี)
ประธานสภาวิศวกร
President

เพื่อใช้ในการรับรองแบบ PV Layout
โครงการ Sainoi PV&BESS Project

Sun J. Amn

5	23/06/2023	As-Built					
4	23/06/2023	Final for Construction					
3	22/06/2023	Issued for Approval					
2	05/06/2023	Issued for Approval					
1	13/08/2022	Issued for Approval					
0	12/04/2022	Issued for Approval					
Rev.	Date	Description					
Owner : ELECTRICITY GENERATING AUTHORITY OF THAILAND							
Contractor : RATCH Group Public Company Limited							
ARCHITECT : -		SUBMITTED		SAI NOI PV AND BESS PROJECT			
ENGINEER : -							
ENGINEER : -		CONCURRED		Document Name SCADA System Configuration Diagram			
DRAWN : -							
CHECKED : -		APPROVED		REPLACING DWG. NO. -	DWG. NO. SNPB-D-GRK- OOO-001	REV NO. 5	SHEET NO. 1/3
DATE							

SAI NOI PV&BESS PROJECT

SCADA SYSTEM CONFIGURATION DIAGRAM

PROJECT NAME :

SAI NOI PV&BESS PROJECT

LOCATION :

EGAT Sai Noi Office at 81
Moo 11 Bangkrual-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



Electricity Generating Authority of Thailand

CONTRACTOR :



RATCH GROUP PUBLIC COMPANY LIMITED

KEY PLAN



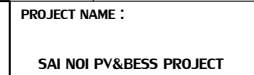
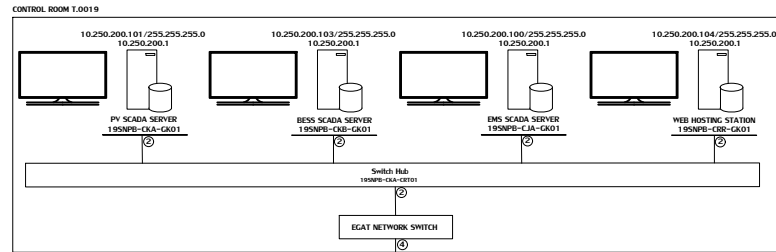
NOTE.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD.
1	FOR APPROVAL	15-08-2023	SAI NOI PV	SAI NOI PV	PLD
2	FOR APPROVAL	15-08-2023	SAI NOI PV	SAI NOI PV	PLD
3	FOR APPROVAL	15-08-2023	SAI NOI PV	SAI NOI PV	PLD
4	FOR CONSTRUCTION	15-08-2023	SAI NOI PV	SAI NOI PV	PLD
5	AS-BUILT	15-08-2023	SAI NOI PV	SAI NOI PV	PLD

DESCRIPTION :

SCADA System Configuration Diagram

DWG BY.	DGN BY.	CHK BY.	APP BY.
DWG NO. SNPB-D-GRK-000-001			
JOB NO.			
SHEET : 1/2	SIZE : A3	SCALE : NTS	



LOCATION :
EGAT Sai Noi Office at 81
Moo 11 Bangkrui-Sai Noi
Road, Sai Noi Sub-district, Sai
Noi District Nonthaburi
Province, Thailand.

OWNER :



Electricity Generating Authority of Thailand

CONTRACTOR :

 **RATCH**
GROUP

RATCH GROUP PUBLIC COMPANY LIMITED

KEY PLAN

The key plan shows the layout of the building, including the entrance, exit, and various rooms. The entrance is located on the left side of the building, and the exit is located on the right side. The building has a complex shape with several rooms of varying sizes. The rooms are labeled with numbers 1 through 10. The entrance is labeled 'ENTRANCE' and the exit is labeled 'EXIT'.

NOTE.

REV	DESCRIPTION	DATE	DRW.	CHK.	APVD
1	FOR APPROVAL	15-APR-2008	SABARWAL P	SABARWAL P	FLOREN
2	FOR APPROVAL	20-APR-2008	SABARWAL P	DANW	DANW
3	FOR APPROVAL	22-APR-2008	SABARWAL P	DANW	DANW
4	FOR CONSTRUCTION	23-APR-2008	SABARWAL P	DANW	DANW
5	AS-BUILT	15-MAY-2008	SABARWAL P	DANW	DANW

DESCRIPTION :

SCADA System Configuration Diagram

DWG BY.	DGN BY.	CHK BY.	APP BY.
DWG NO. SNPB-D-GRK-000-001			
JOB NO.			
SHEET : 2/2	SIZE : A3	SCALE :NTS	